

Beyond Generation: Toward Physical, Agentic, and Actionable World Models

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<https://liuziwei7.github.io>



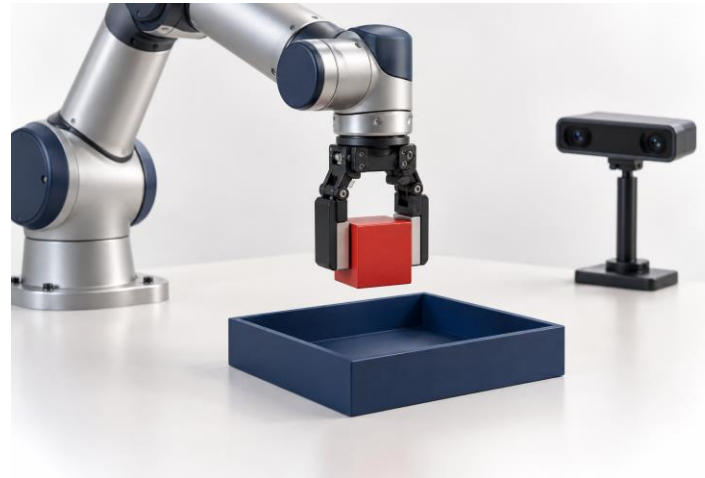
Why Do We Need World Models?

Autonomous Driving



Anticipate the consequences of a driving decision.

Robot Manipulation



Evaluate candidate plans before committing to execution.

Interactive Environments



Create varied scenarios for training and testing model behavior.

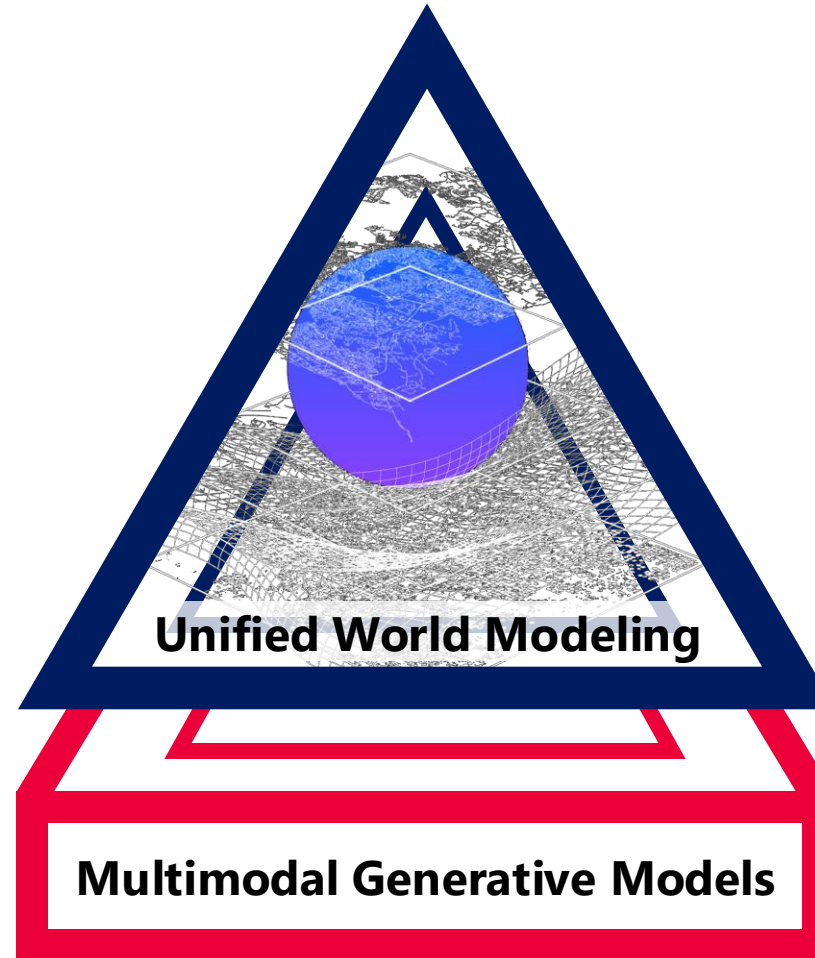
World model is a learned representation of an environment that predicts how it evolves in response to actions.

Challenges

Be Physical

How to Model Physical Properties

Be Agentic
How to Understand
the Physical World

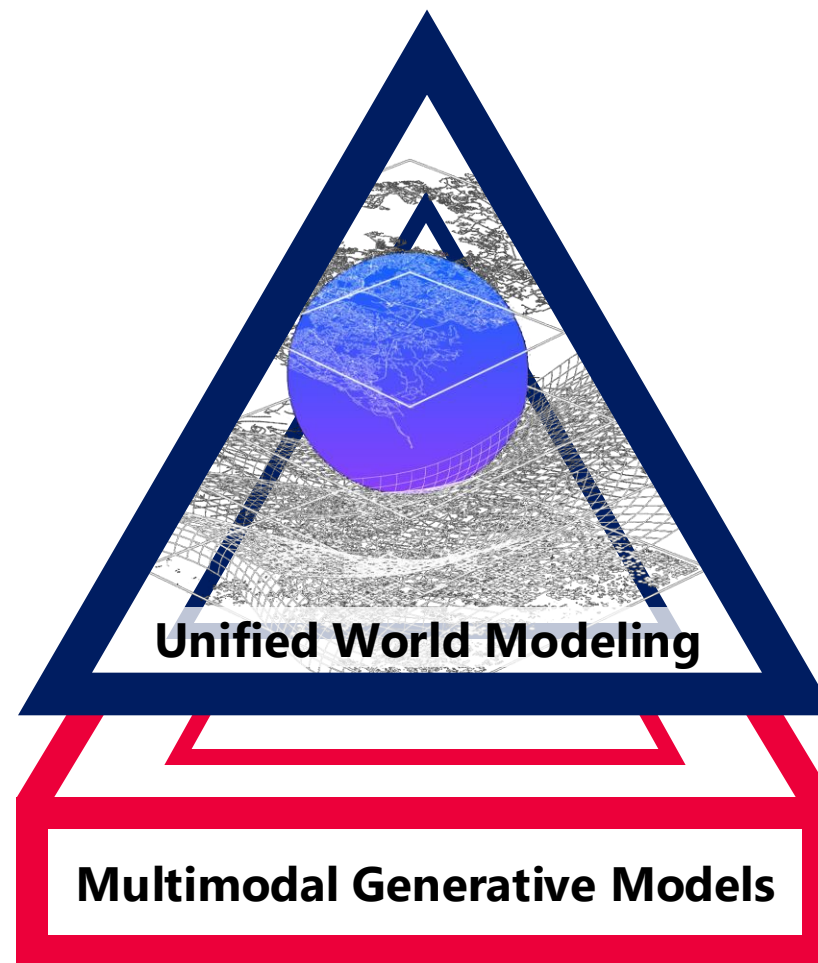


Be Actionable
How to Interact with
the Physical World

Be Physical

How to Model Physical Properties

Be Agentic
How to Understand
the Physical World



Be Actionable
How to Interact with
the Physical World

Be Physical: PhysX-Anything

 [Ziangcao0312/PhysX-Anything](https://github.com/Ziangcao0312/PhysX-Anything)

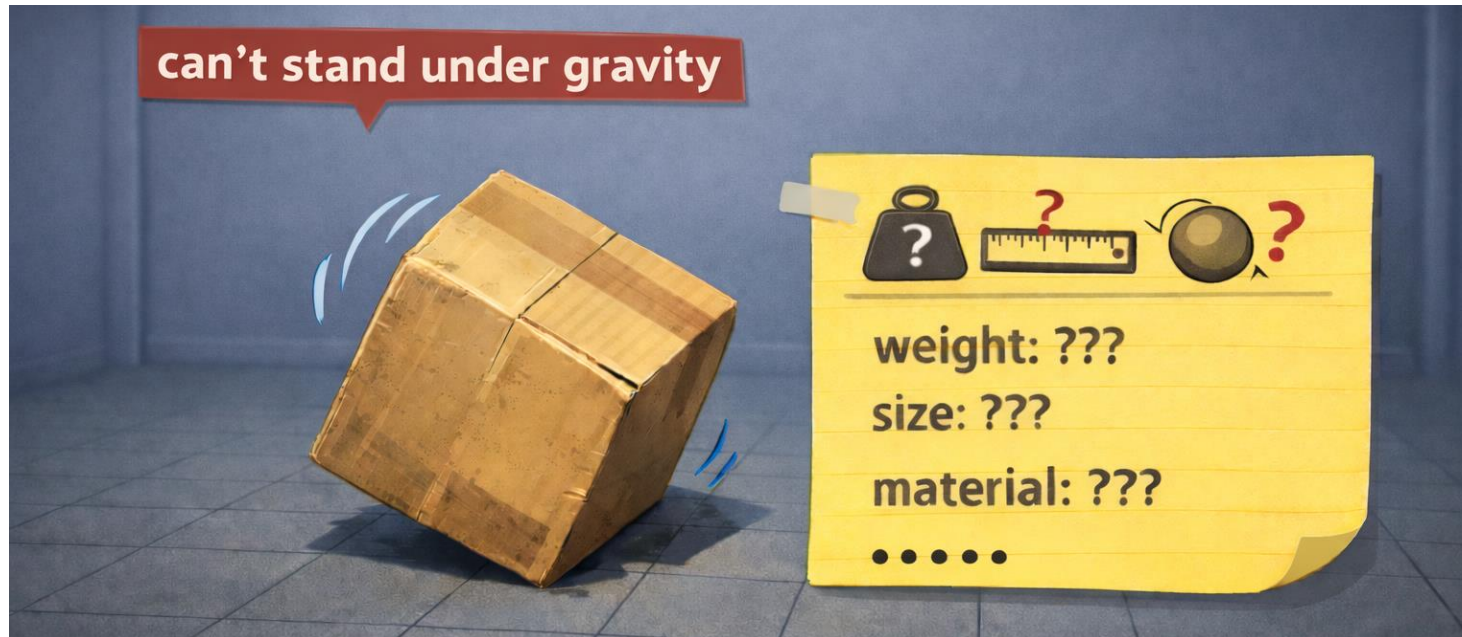
PhysX-Anything: Simulation-Ready Physical 3D Assets from Single Image

Ziang Cao, Fangzhou Hong, Zhaoxi Chen, Liang Pan, Ziwei Liu
CVPR 2026

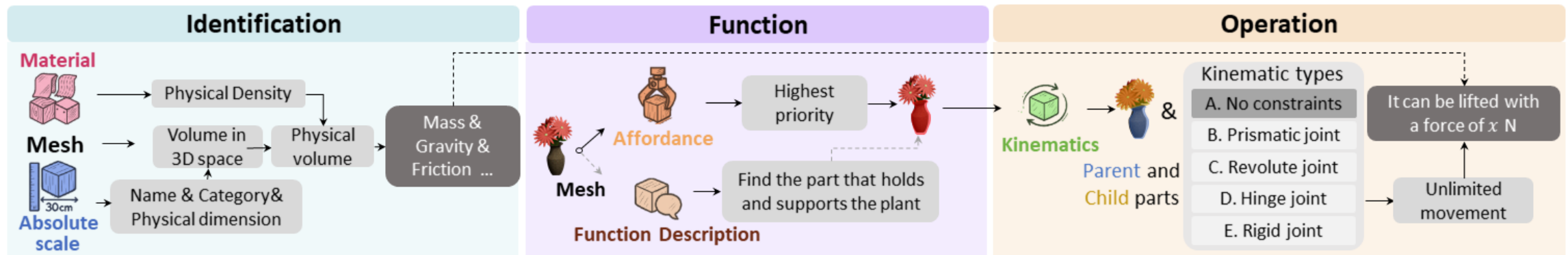
Challenges

■ Modelling of Physical Attributes

- Existing 3D generation primarily emphasizes geometries and textures while neglecting **physical-grounded modeling**, hampering their real-world application in physical domains like simulation and embodied AI.



Definition of Physical Properties



Identification – Determining the basic nature of the object

Absolute scaling and material (Young's modulus, Poisson's ratio, and density)

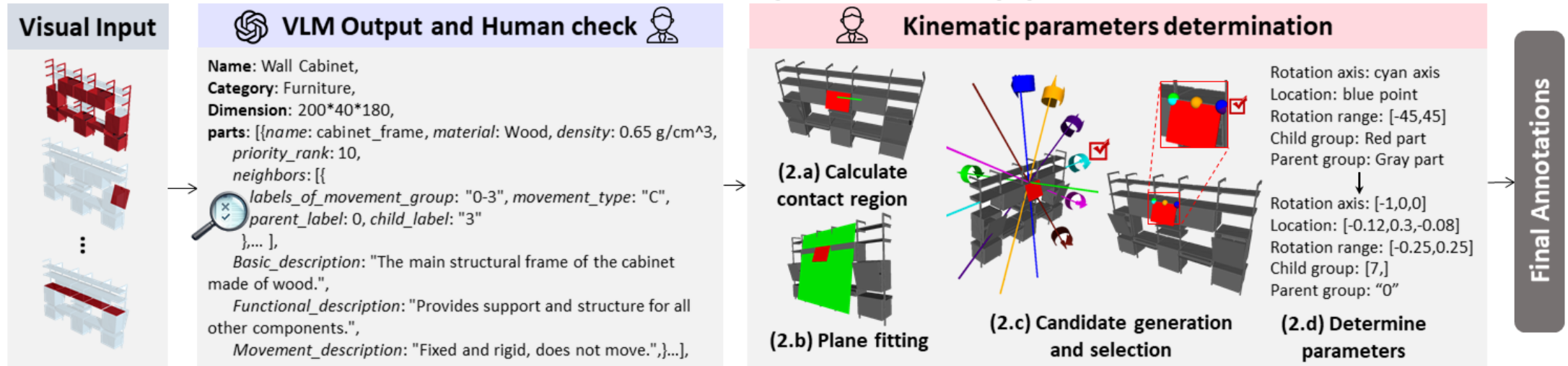
Function – Understanding its potential applications

Functional affordance analysis and function descriptions

Operation – Detailed usage methodologies

Kinematic Parameter

Human-in-the-loop Annotation Pipeline



Preliminary Data Acquisition

GPT-4o

Human Check

Kinematic Parameter Determination

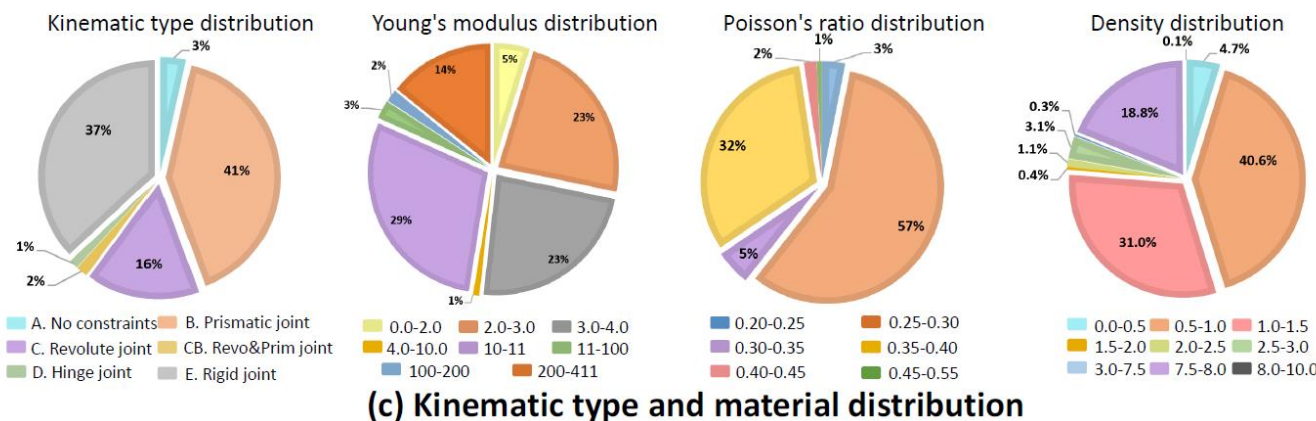
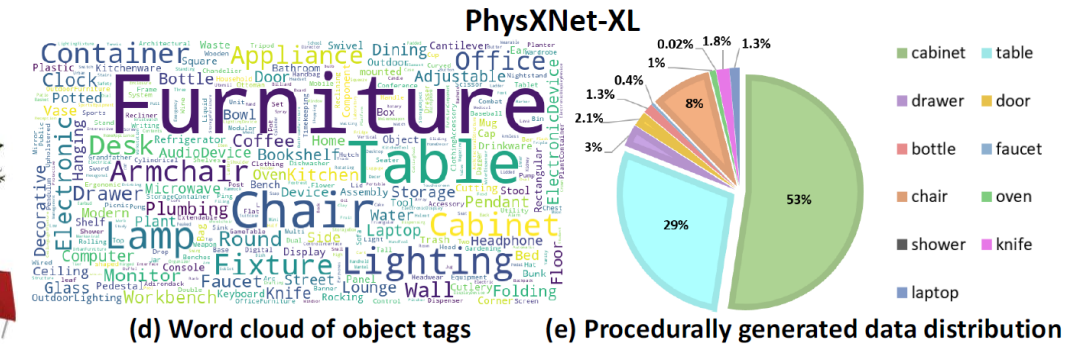
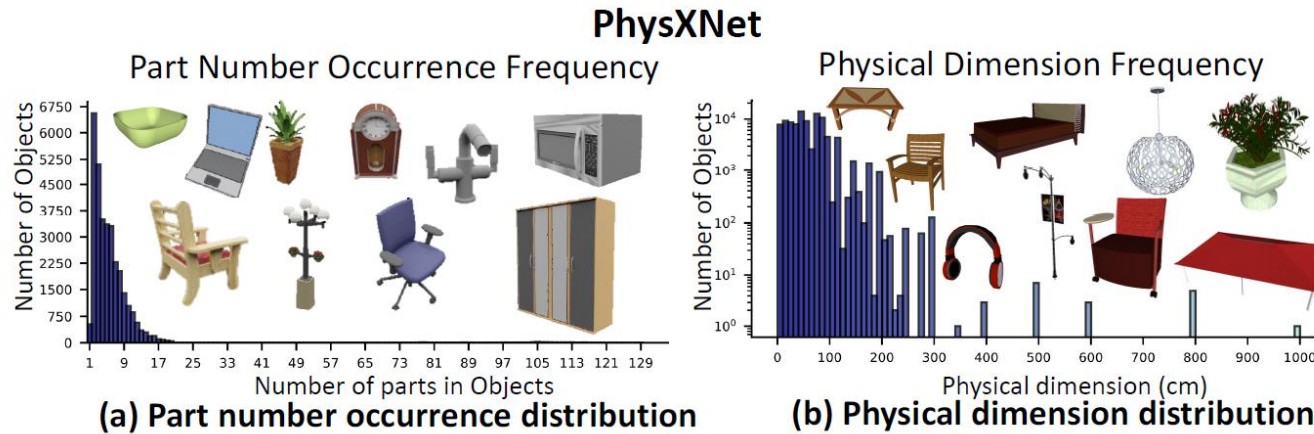
Contact Region

Plane Fitting

Candidate Generation
and Selection

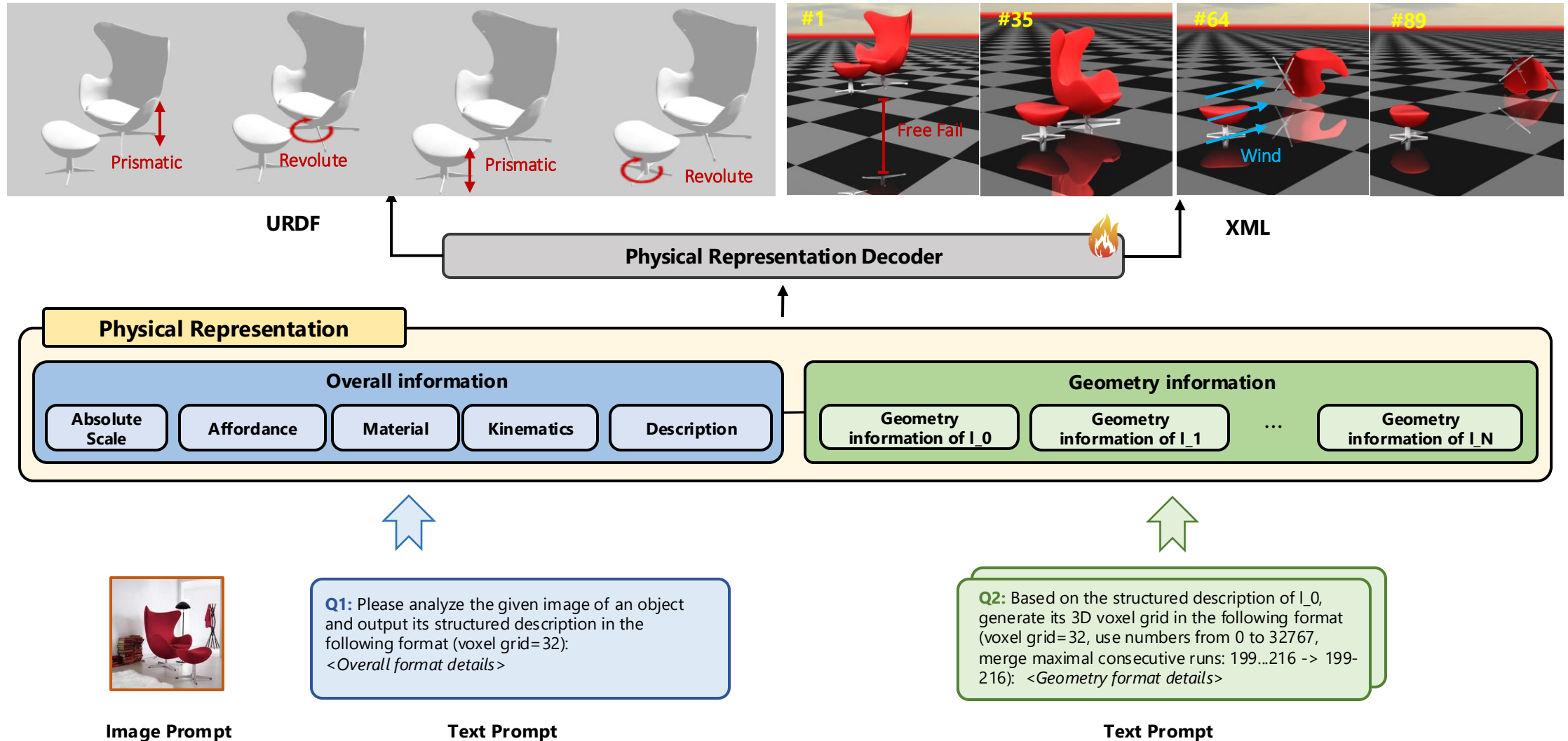
Kinematic Parameter
Determination

PhysXNet & PhysXNet-XL

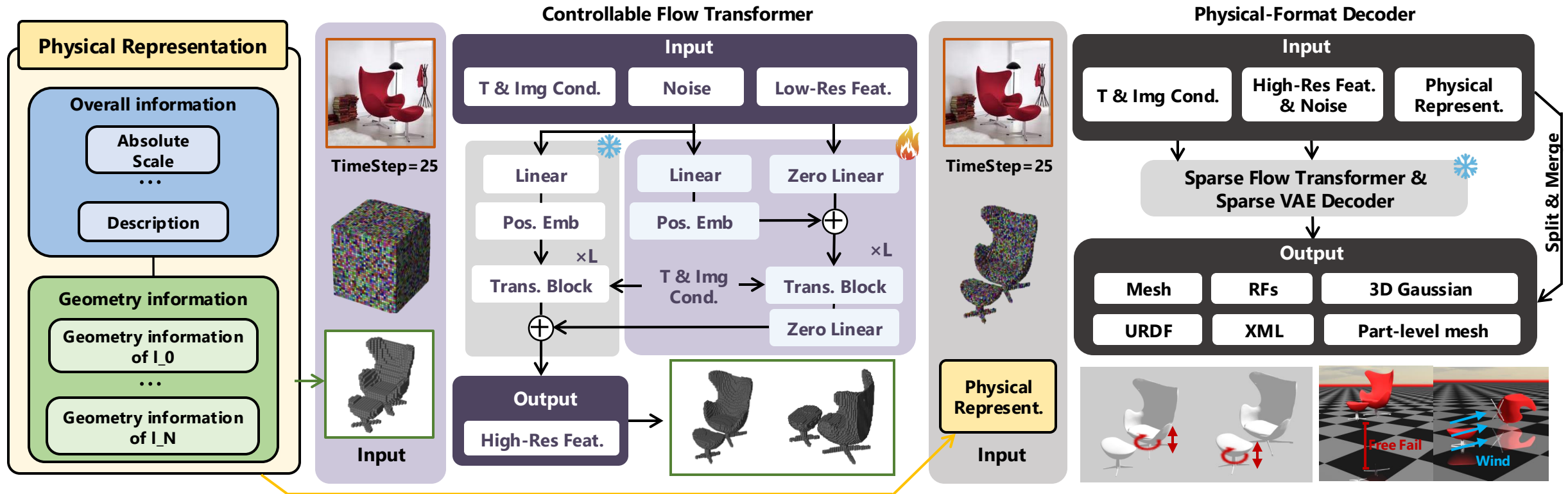


We propose **PhysXNet** and **PhysXNet-XL** – the first comprehensive physical 3D dataset with over **26K** and **6M** richly annotated 3D objects

Key Idea: Interactive 3D Object as Physical Code



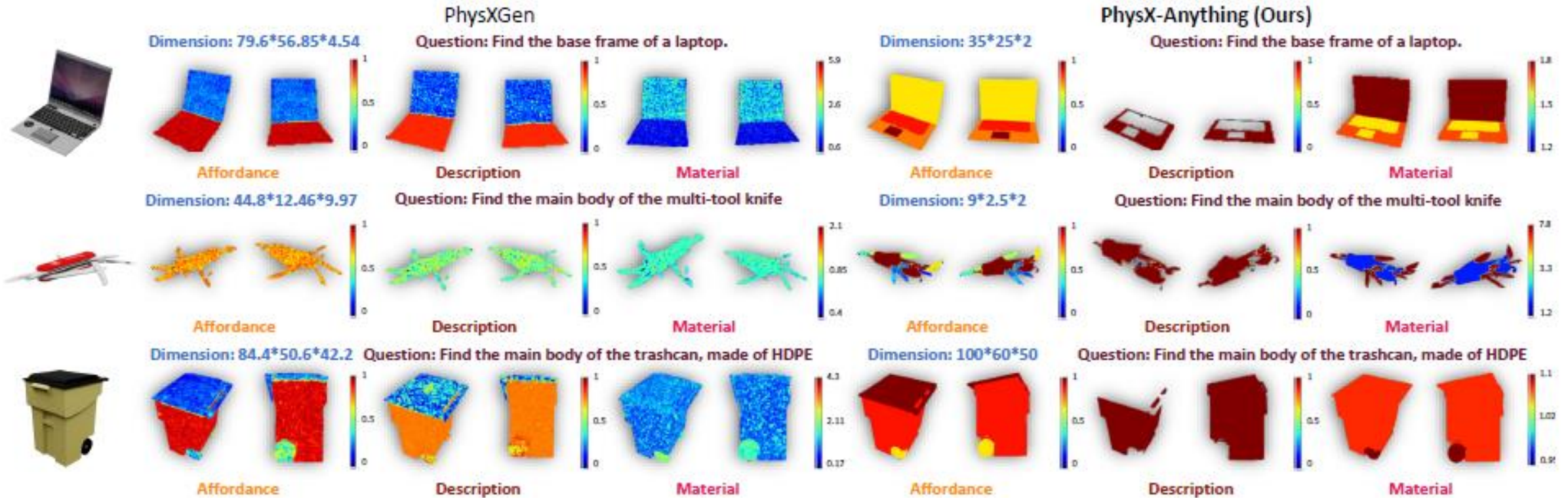
Method: Physical Representation Decoder



Results: Qualitative Comparisons



Results: Qualitative Comparisons



Results: Quantitative Comparisons

Evaluations on PhysX-Mobility

Methods	Geometry			Physical Attributes				
	PSNR $\uparrow$	CD $\downarrow$	F-score $\uparrow$	Absolute scale $\downarrow$	Material $\uparrow$	Affordance $\uparrow$	Kinematic parameters (VLM) $\uparrow$	Description $\uparrow$
URDFormer [11]	7.97	48.44	43.81	–	–	–	0.31	–
Articulate-Anything [16]	16.90	17.01	67.35	–	–	–	0.65	–
PhysXGen [3]	20.33	14.55	76.3	43.44	6.29	9.75	0.71	12.89
PhysX-Anything (Ours)	20.35	14.43	77.50	0.30	17.52	14.28	0.83	19.36

In-the-wild Evaluations

Methods	Geometry (Human) $\uparrow$	Physical Attributes (Human)				
		Absolute scale $\uparrow$	Material $\uparrow$	Affordance $\uparrow$	Kinematic parameters $\uparrow$	Description $\uparrow$
URDFormer [11]	0.21	–	–	–	0.23	–
Articulate-Anything [16]	0.53	–	–	–	0.37	–
PhysXGen [3]	0.61	0.48	0.43	0.34	0.32	0.33
PhysX-Anything (Ours)	0.98	0.95	0.84	0.94	0.98	0.96

Results: Simulated Objects

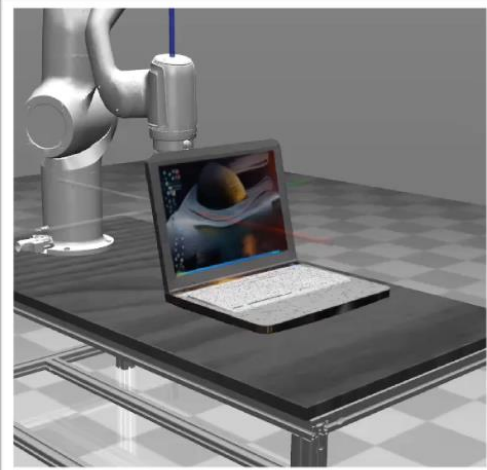


Results: Quantitative Comparisons

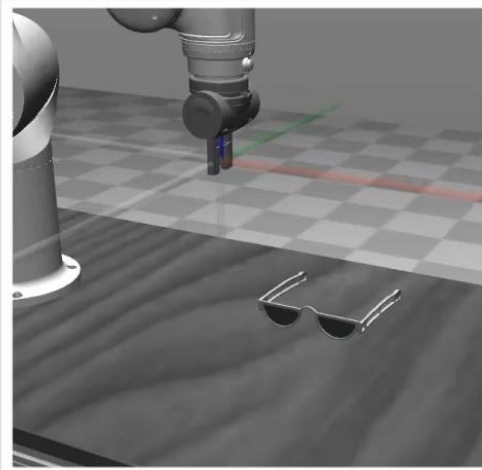


Results: Simulated Interactions

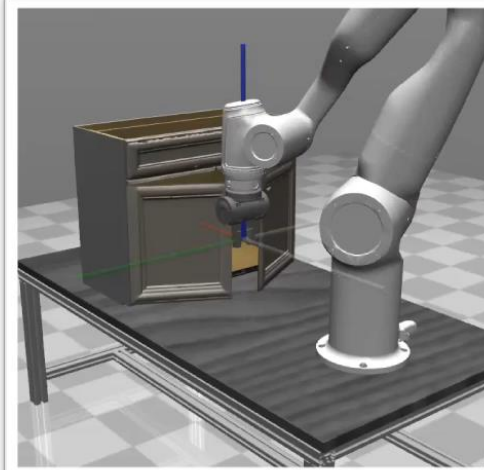
Laptop Closing



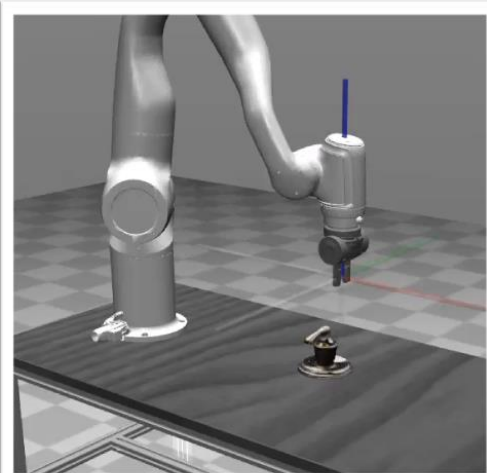
Eyeglass Temple Folding



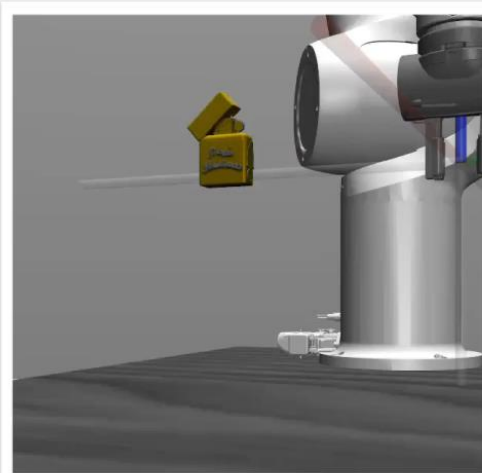
Door Opening and Closing



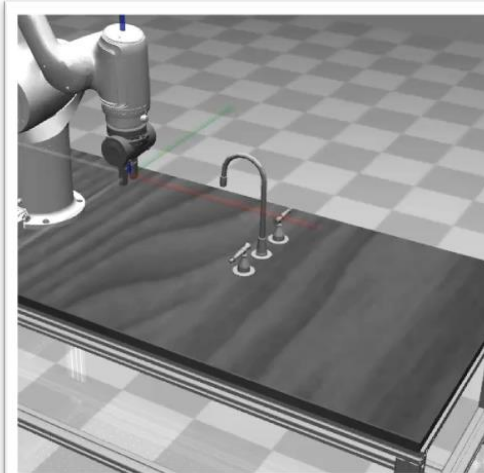
Handle Manipulation



Lighter Snapping Open



Faucet Switch Manipulation



Be Physical: PhysX-Omni

 [Ziangcao0312/PhysX-Omni](https://github.com/Ziangcao0312/PhysX-Omni)

PhysX-Omni: Unified Simulation-Ready Physical 3D Generation for Rigid, Deformable, and Articulated Objects

Ziang Cao, Yinghao Liu, Haitian Li, Runmao Yao, Fangzhou Hong, Zhaoxi Chen, Liang Pan, Ziwei Liu

Challenges

- Limited Coverage of Physical Object Types
 - Most 3D generation methods focus on geometry and appearance. Even simulation-ready approaches such as PhysX-Anything primarily target articulated objects, offering **limited support** for rigid and deformable objects.

Most 3D Generation Methods



Geometry

Appearance



Fabric deformation

Unknown kinematics

Unknown size & position

Category-Specific Sim-Ready Methods



Rigid

Articulated

Deformable

Challenges

■ Scarcity of Physical 3D Data

- High-quality 3D datasets with comprehensive physical annotations remain scarce. Their **limited scale and diversity** constrain generation quality and generalization.

■ Lack of Ground-Truth Benchmarks

- High-quality 3D datasets with comprehensive physical annotations remain scarce. Their **limited scale and diversity** constrain generation quality and generalization.

Conventional 3D Datasets



Geometry ✓ Appearance ✓

Material ? Kinematics ? Scale ?

Physical 3D Datasets



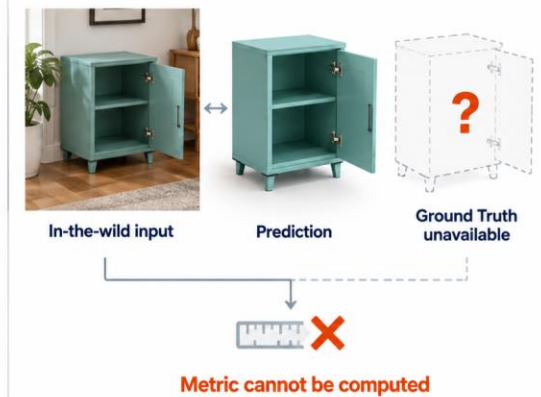
Material ✓ Kinematics ✓ Scale ✓

Limited object diversity

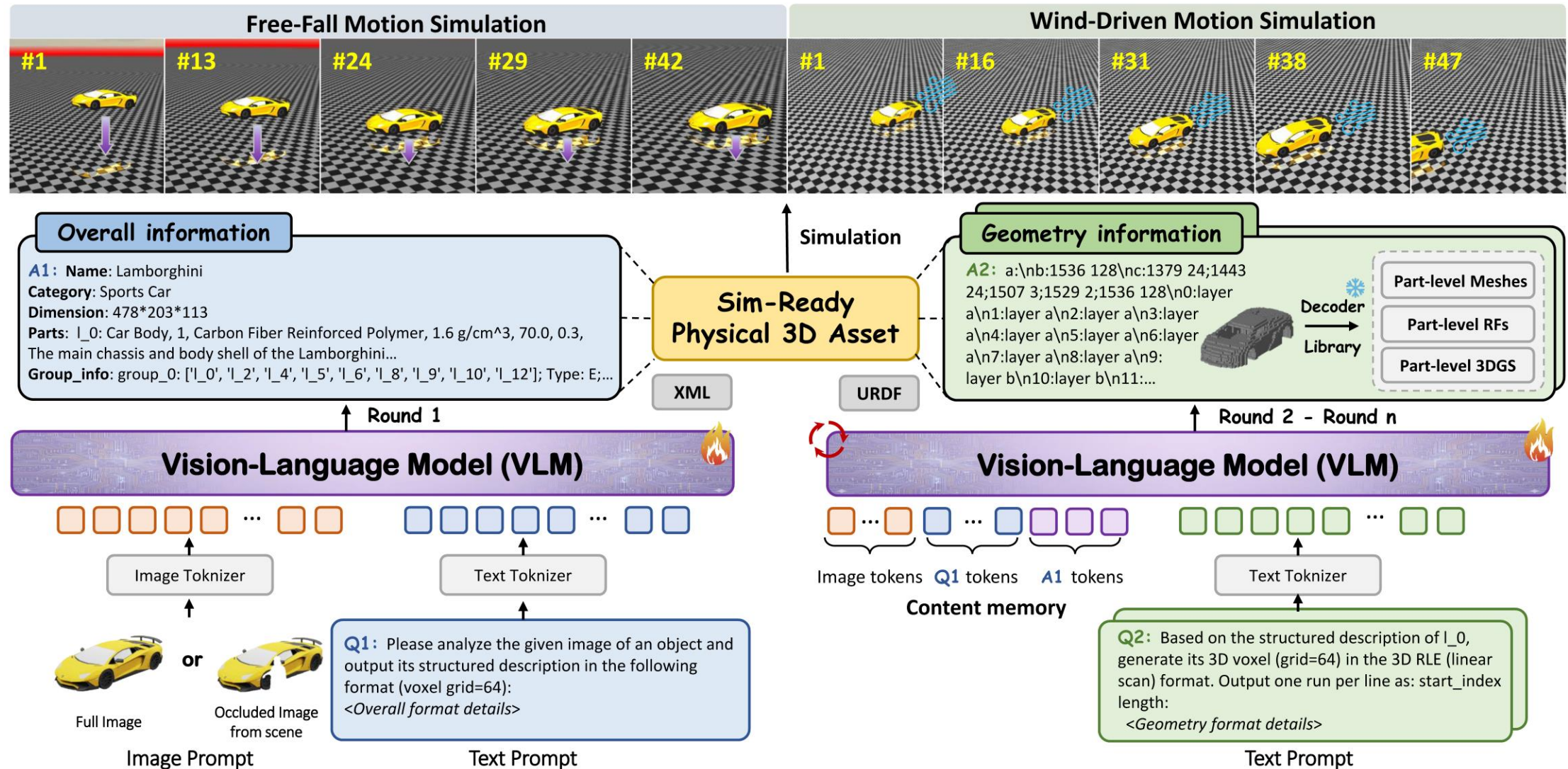
With Ground Truth



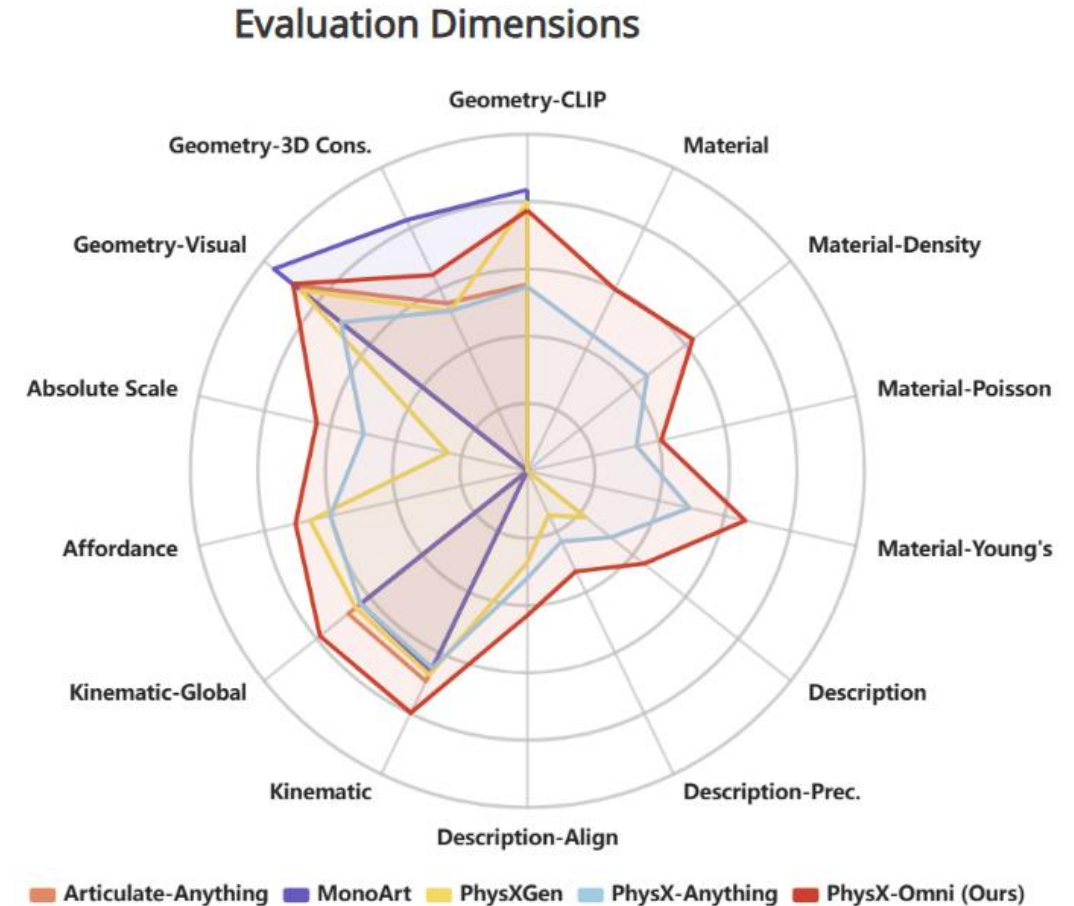
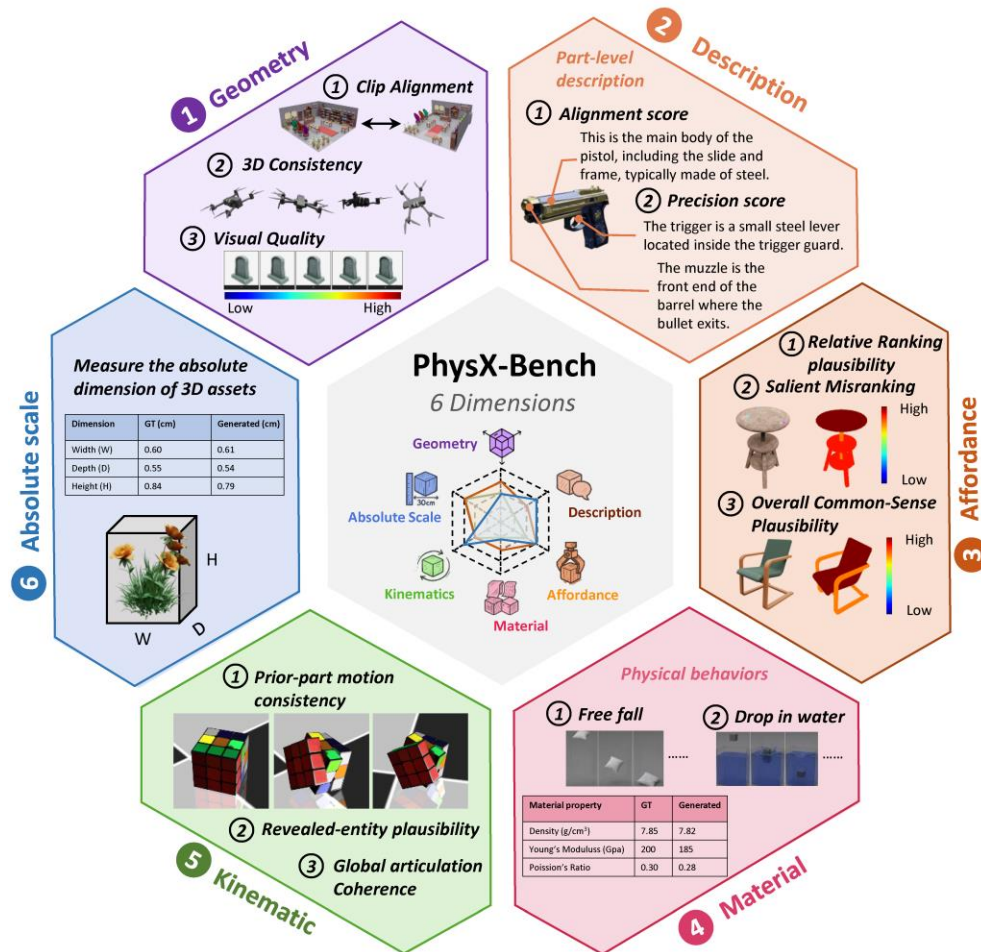
Without Ground Truth



Key Idea: Diverse 3D Object as Unified Physical Code

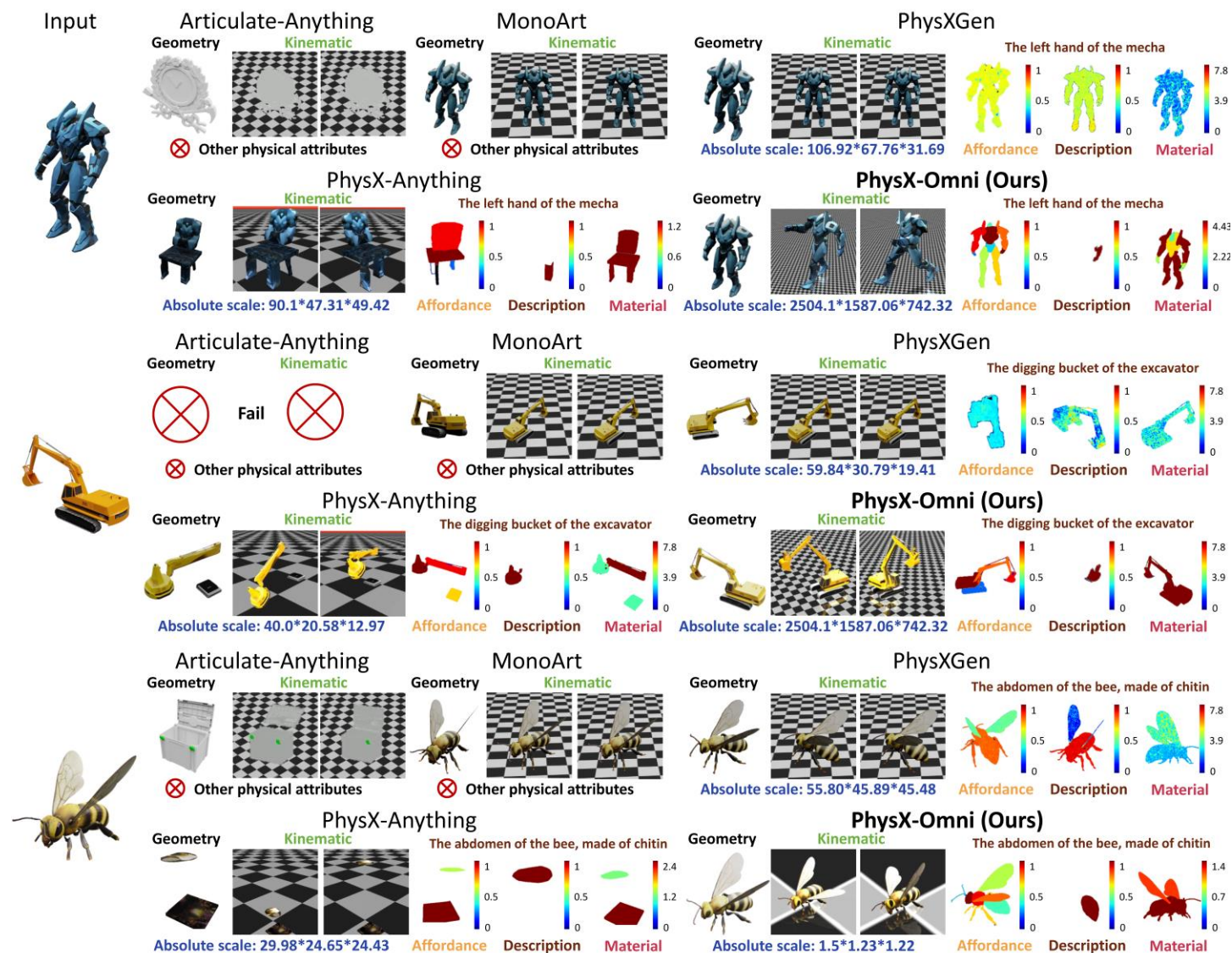


PhysX-Bench

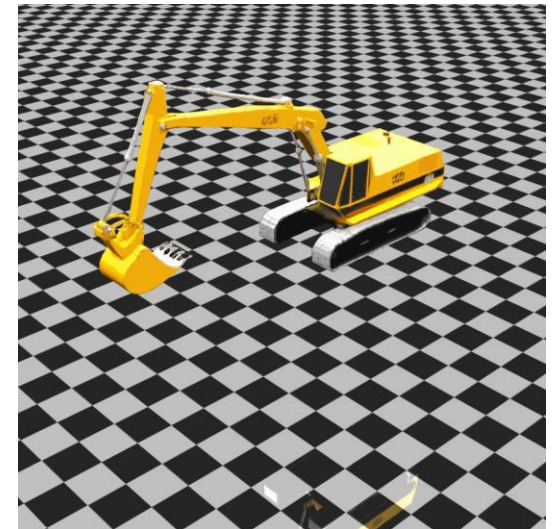
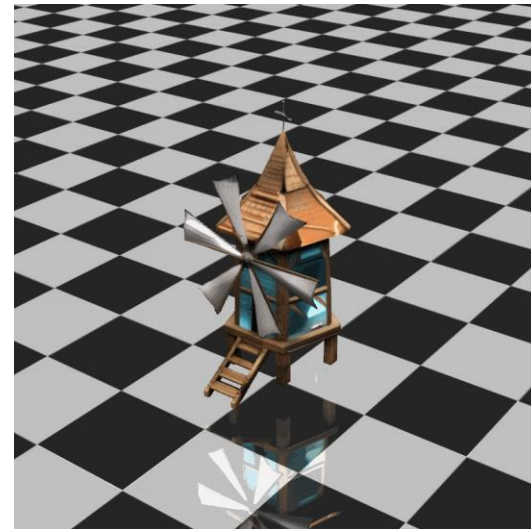
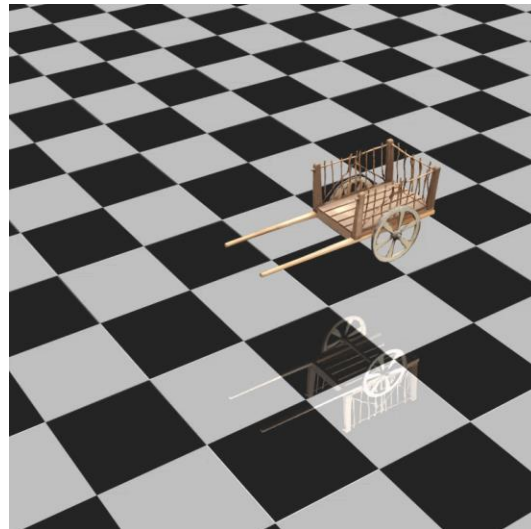
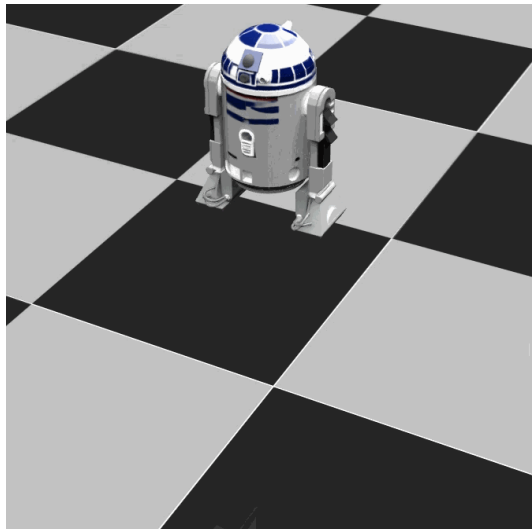
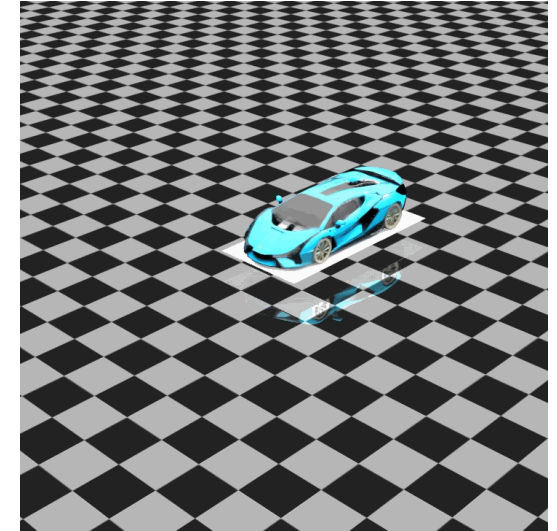
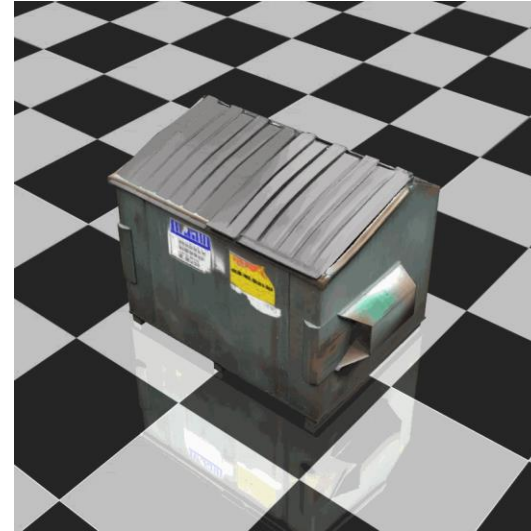
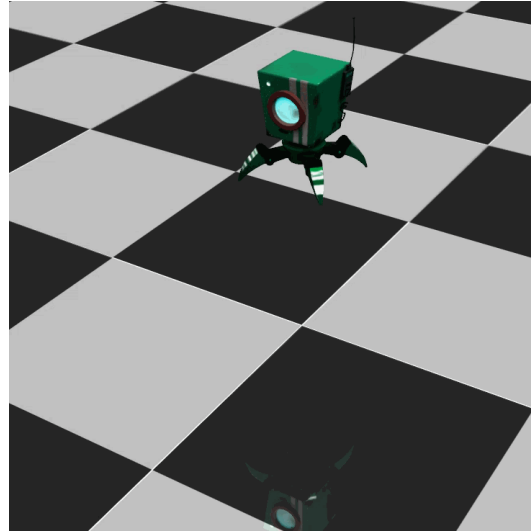


It consists of **six key dimensions** for comprehensively evaluating 3D structure, appearance, fundamental physical attributes, and understanding

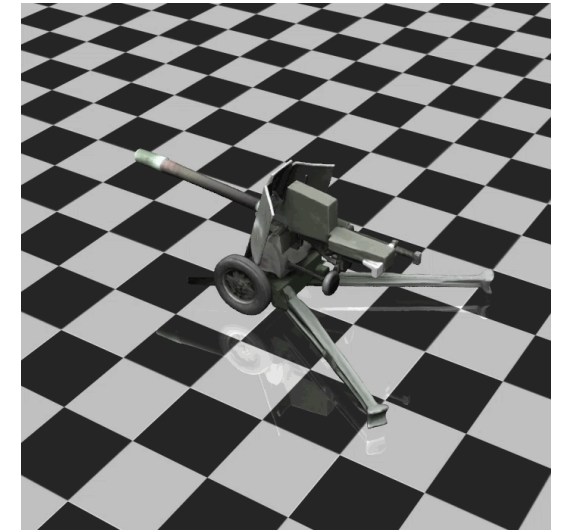
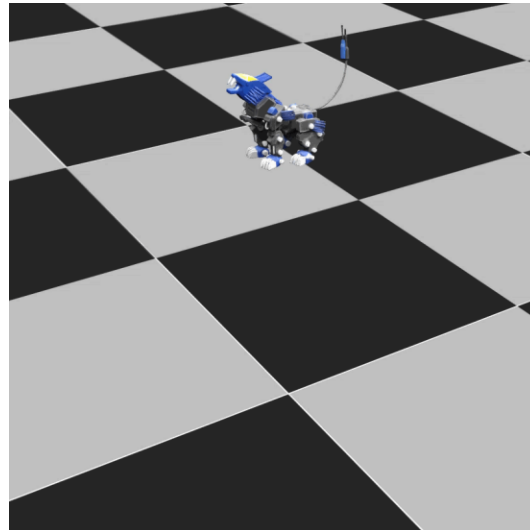
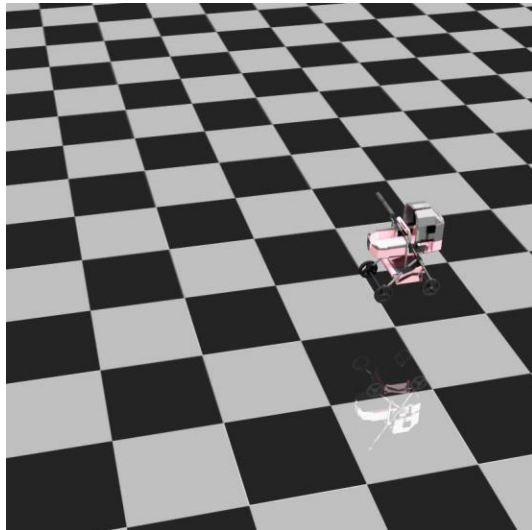
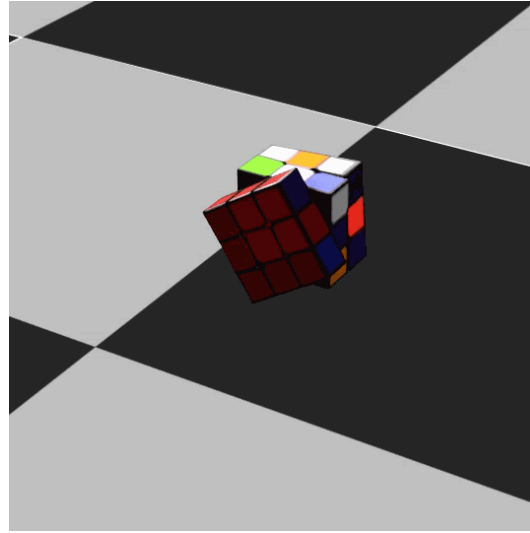
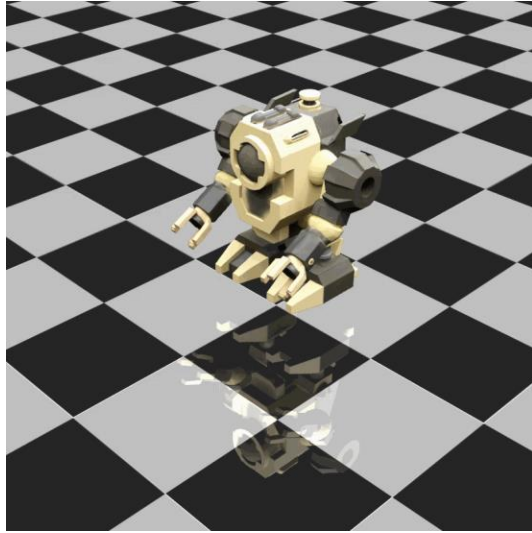
Results: Qualitative Comparisons



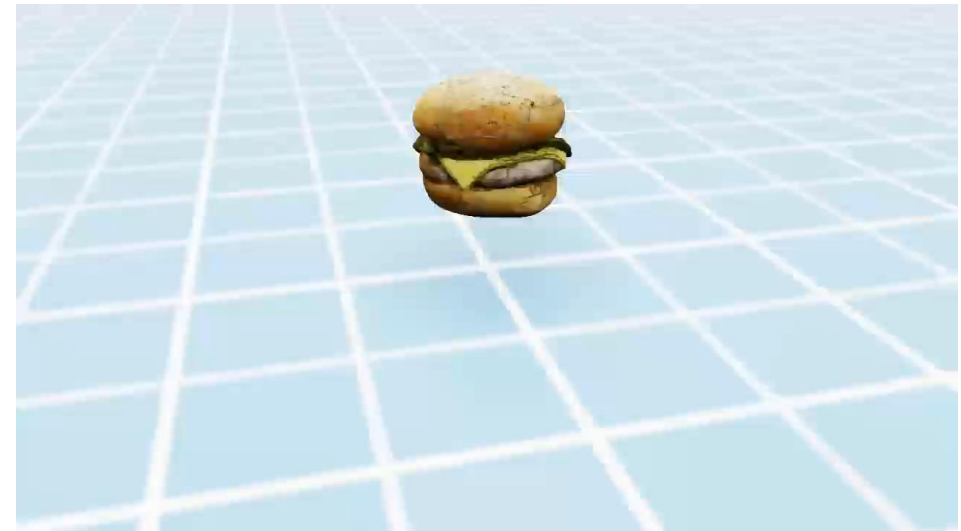
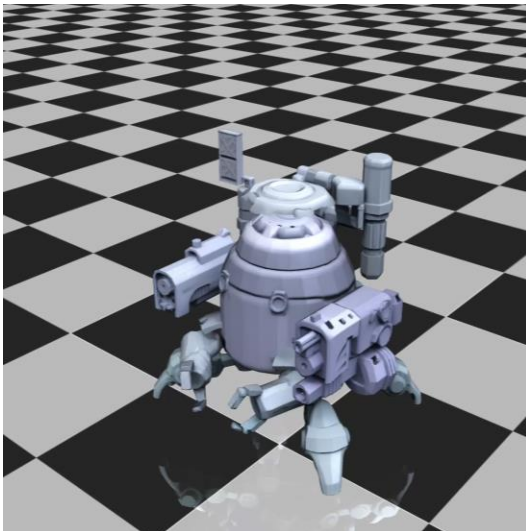
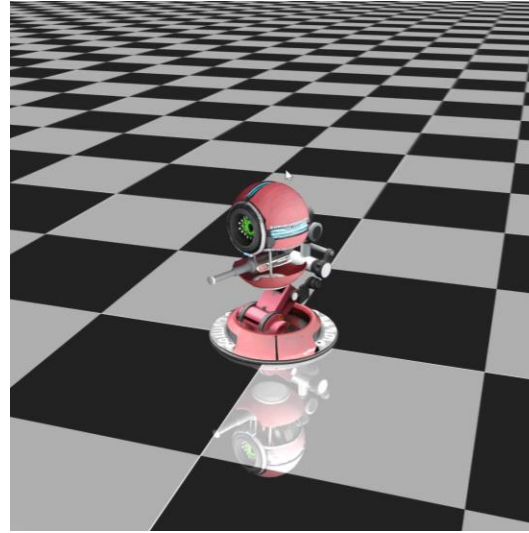
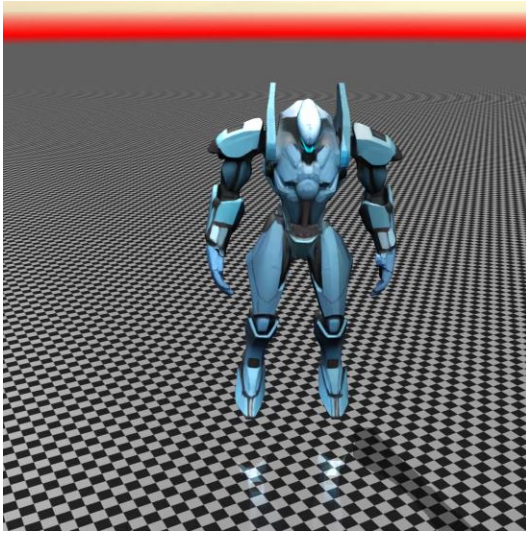
Results: Simulated Objects



Results: Simulated Objects



Results: Simulated Objects



Results: Simulated Objects

Close the Turntable Lid



Move the Wheelchair



Open Toilet Cover



Close the Cabinet Door

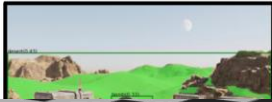


Application: Simulated Scene

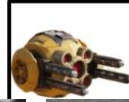
Inputs



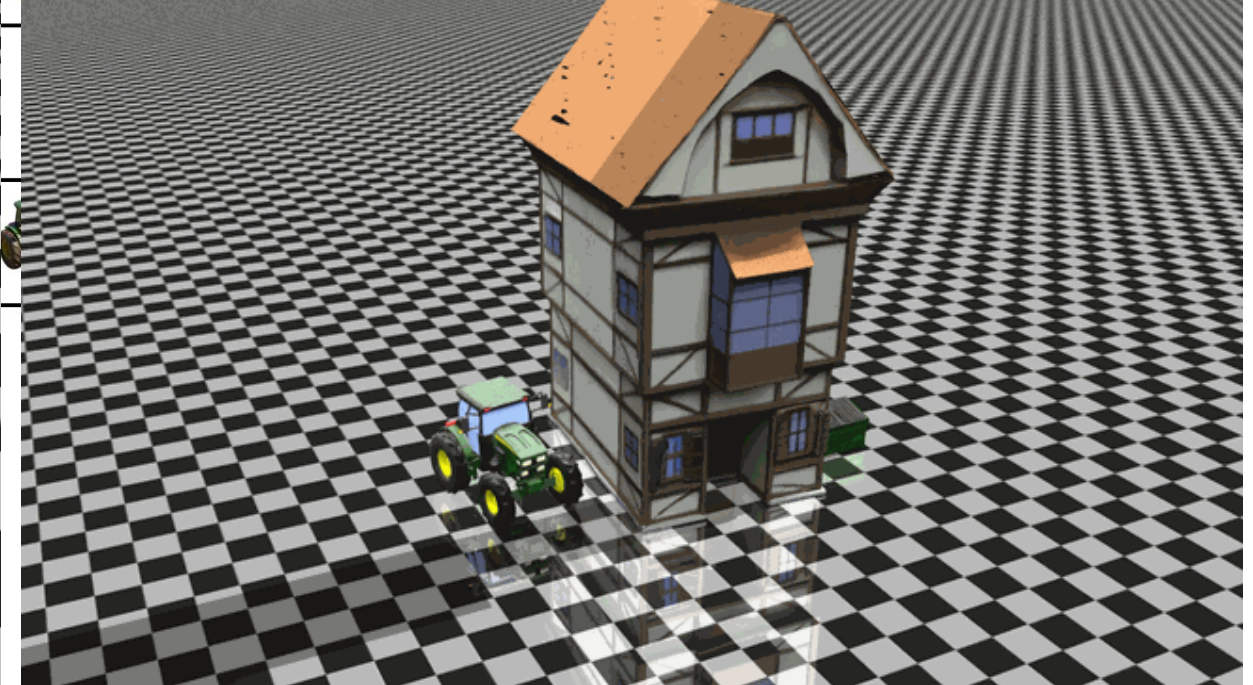
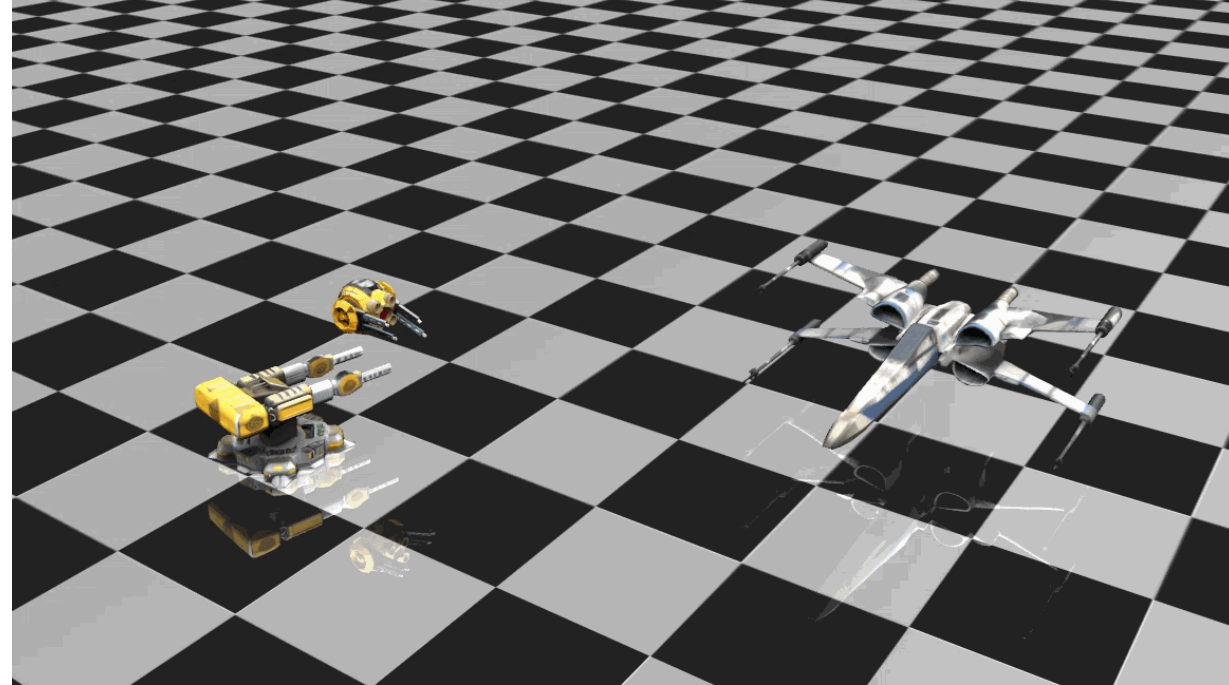
2D image to 3D Layout



Assets in Scene



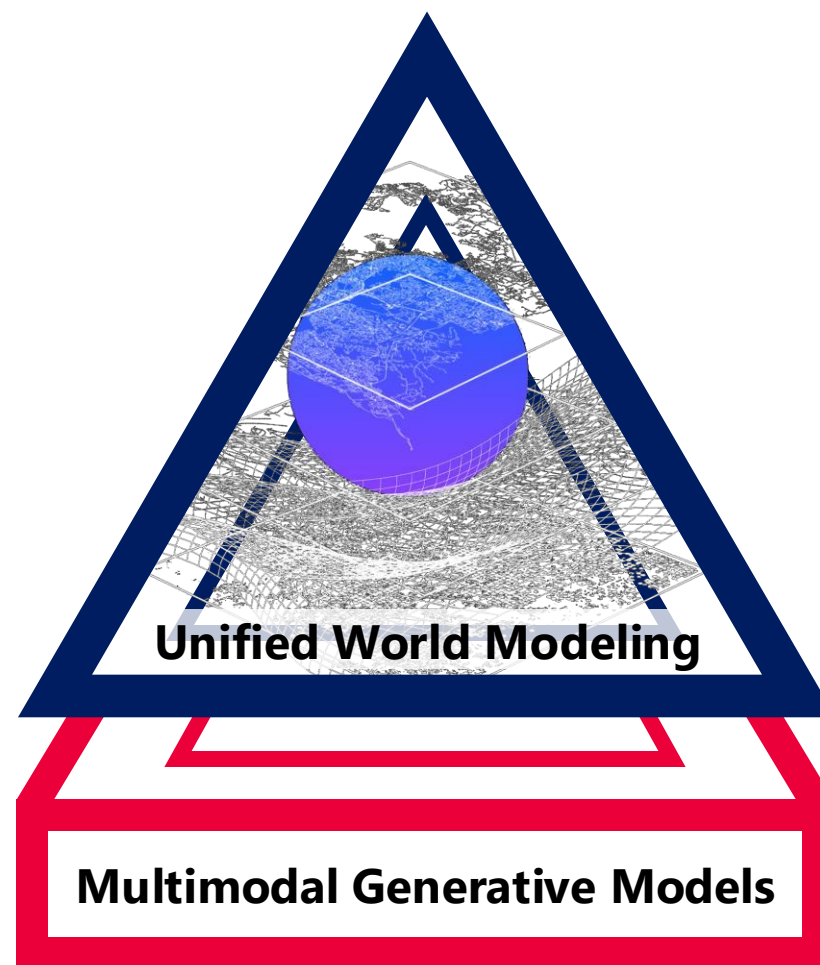
Generated Sim-Ready Scenes



Be Physical

How to Model Physical Properties

Be Agentic
How to Understand
the Physical World



Be Actionable
How to Interact with
the Physical World

Motivation: S-Agent

How can an agent build a coherent 3D scene from partial views?

Limits of isolated visual observations

VLMs learn mainly from 2D image-text data. Semantic recognition does not reliably preserve metric 3D geometry.

Each frame reveals only part of a scene. Independent predictions can lose object states and evidence across views and time.

Accumulate persistent spatial evidence

Let the VLM plan what to inspect or measure, while spatial tools and experts recover the geometry.

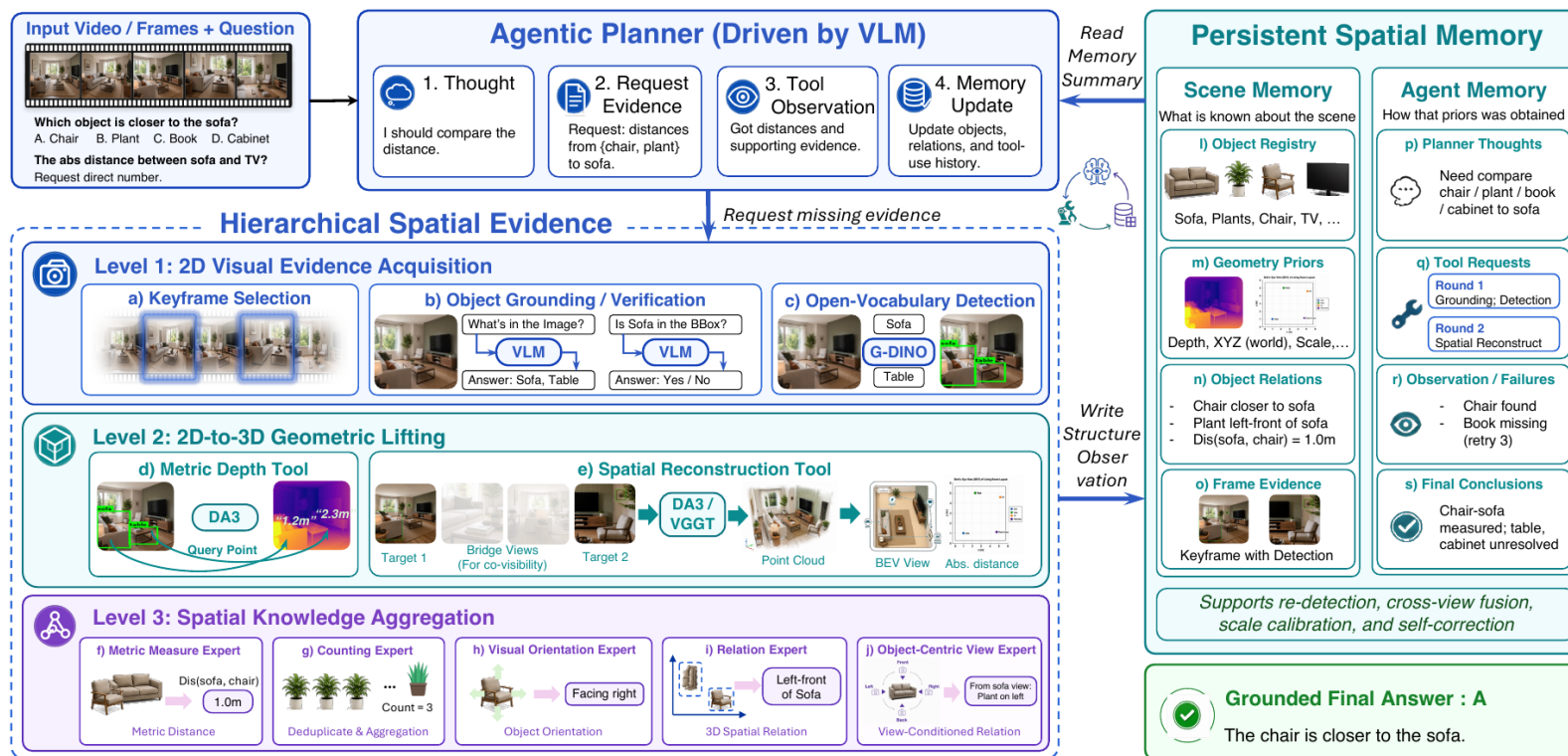
Scene memory preserves object states.
Agent memory retains tool observations and reasoning for reuse in later steps.

Spatial Tool-Use Elicits Reasoning for Spatial Intelligence

Yalun Dai et al., 2026

Key Idea: Spatial Tools with Persistent Memory

A VLM chooses the evidence. Spatial tools and memory ground its reasoning.



VLM planning

The VLM asks for missing evidence, inspects tool outputs, and decides when it can answer.

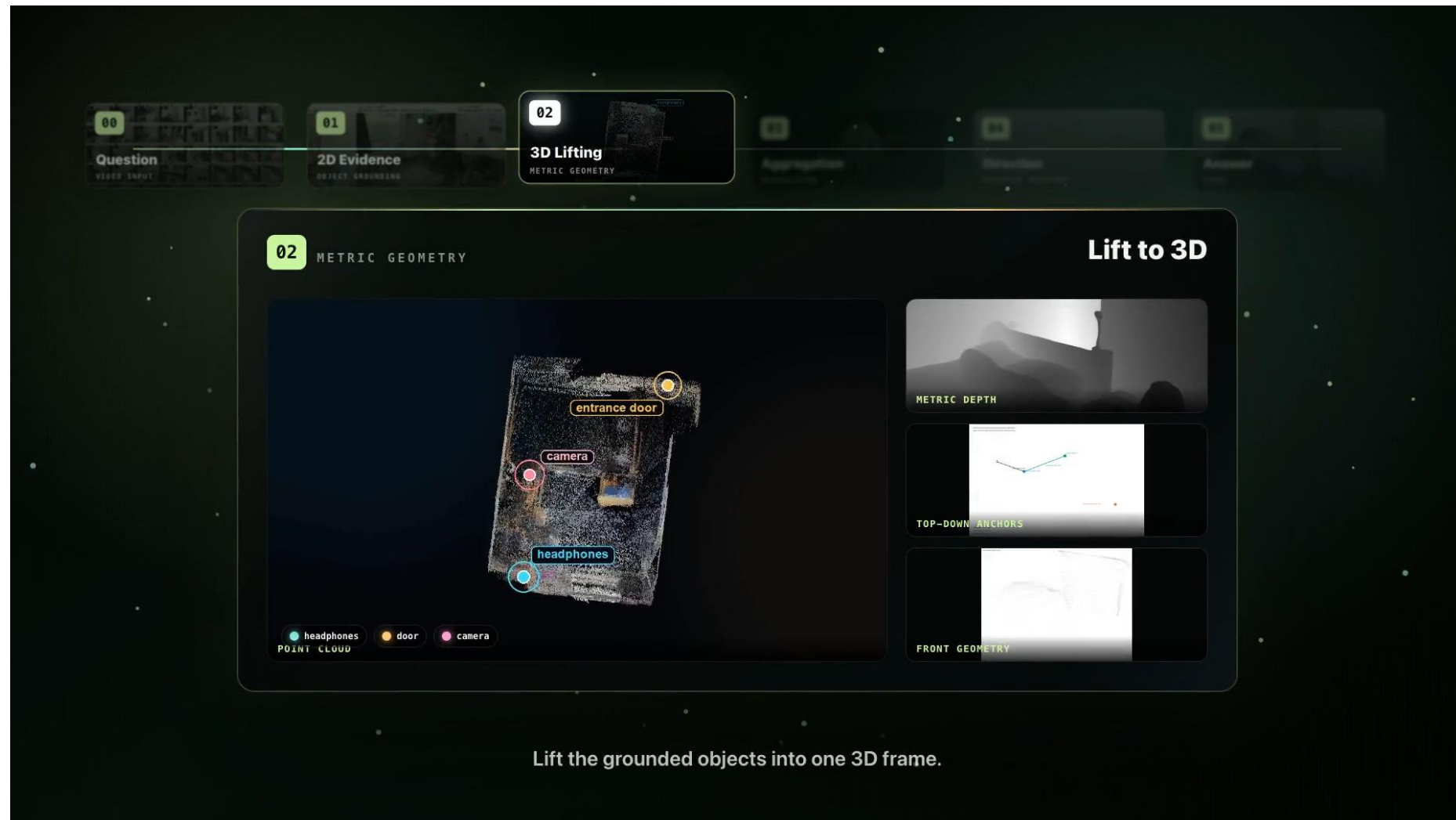
Hierarchical spatial tools

2D grounding and 3D lifting feed spatial experts that compute counts, distances and relations.

Dual persistent memory

Scene memory merges object evidence. Agent memory keeps the reasoning and tool history.

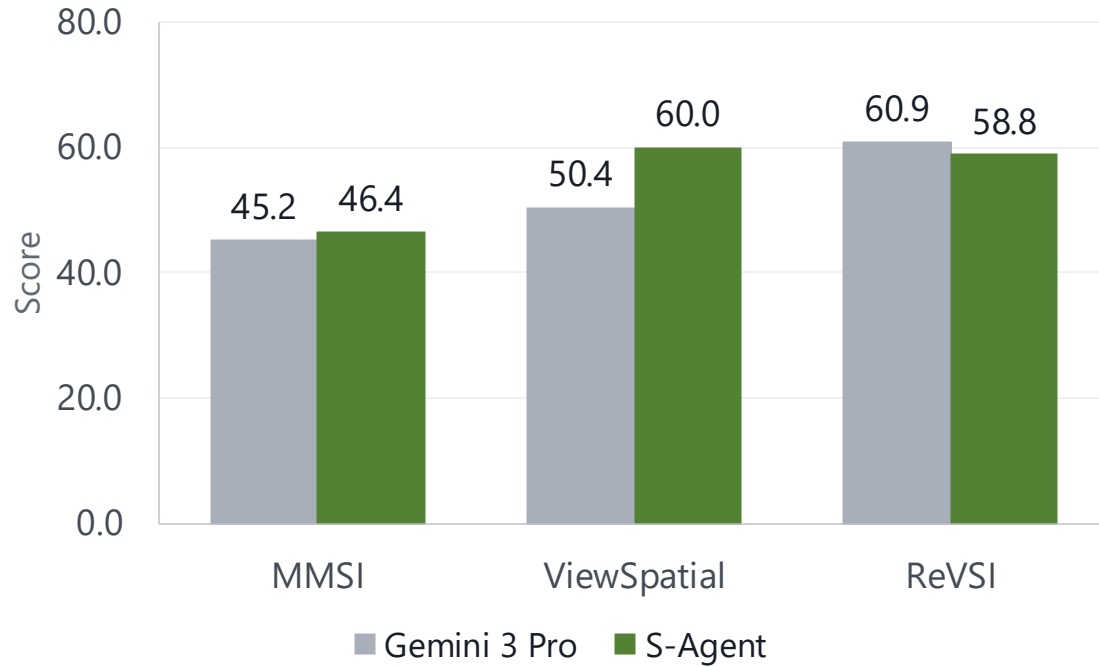
Demo: S-Agent



Click to play full screen (55 s)

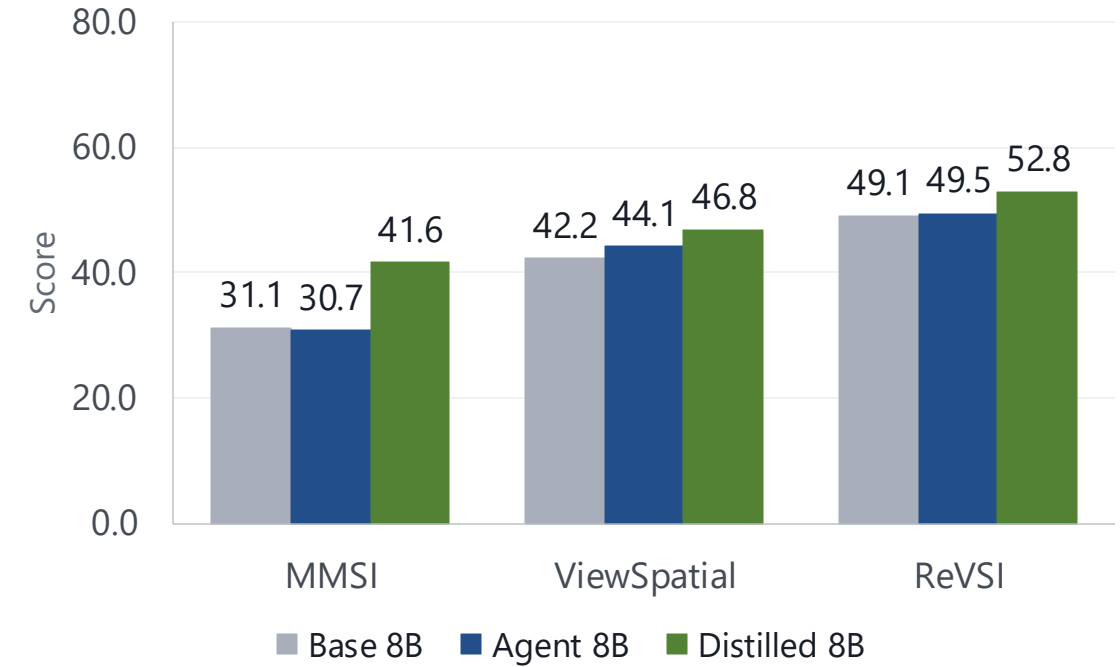
Results: Spatial Reasoning and Distillation

Zero-shot spatial reasoning



+9.6 points on ViewSpatial
versus Gemini 3 Pro

Distilling Qwen3-VL-8B on S-300K



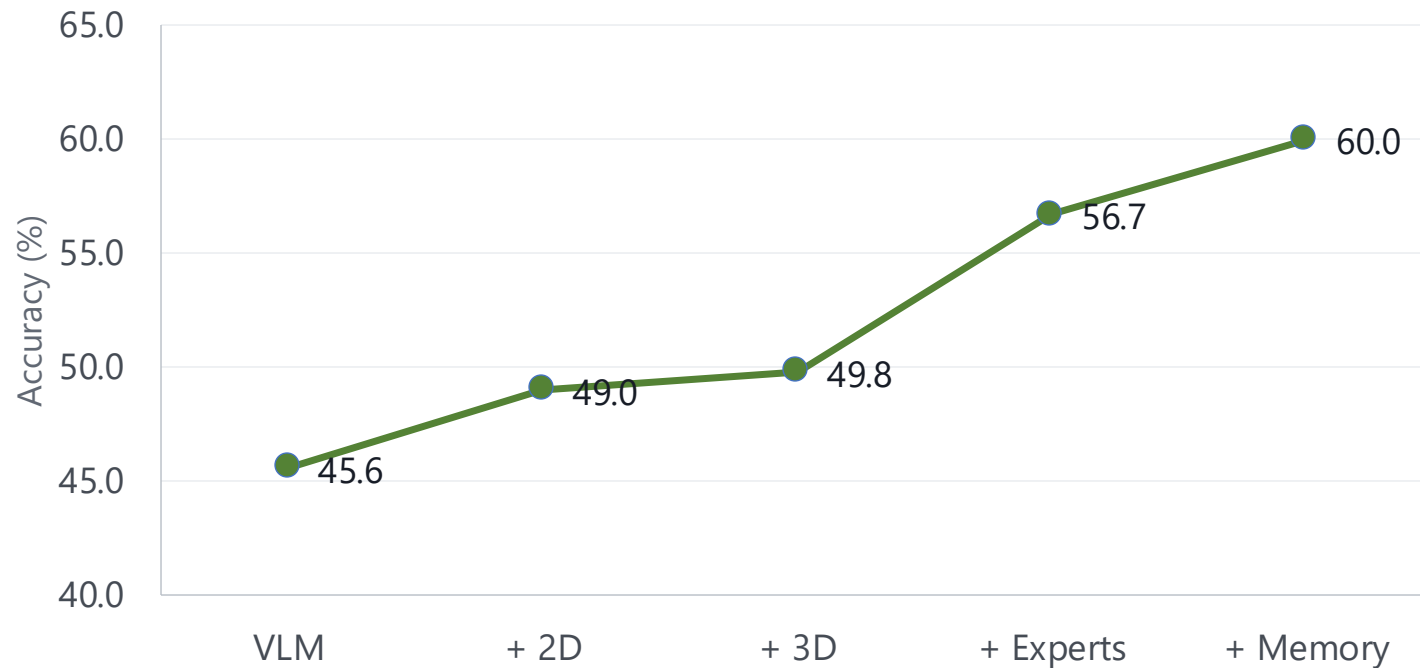
+10.5 points on MMSI
versus the base 8B model

Benchmark scores follow each official evaluation protocol. Higher is better.

Ablations: How Spatial Evidence Adds Value

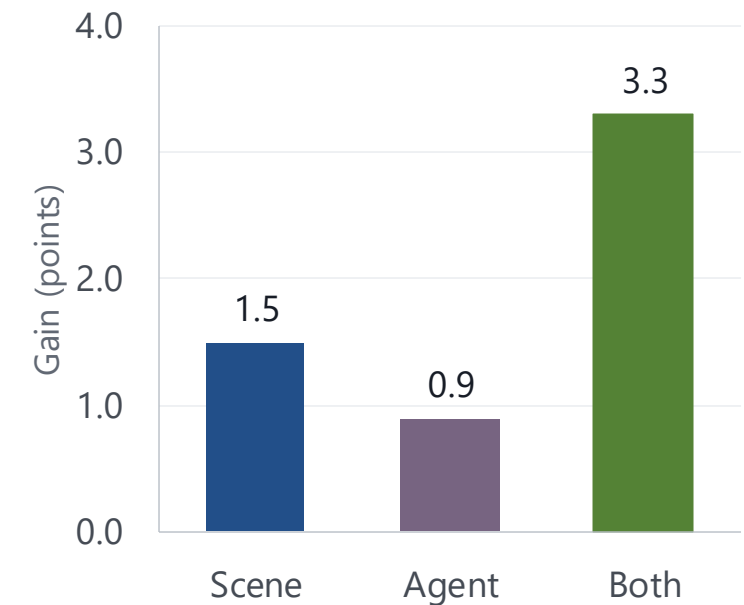
GPT-5.4 planner on ViewSpatial-Bench

Cumulative spatial capabilities



+6.9 points from spatial experts
49.8% to 56.7%

Memory gain over spatial hierarchy



Both memories: 60.0%
Hierarchy alone: 56.7%

Motivation: S-Agent 2.0

Why can correct 3D measurements still lead to a wrong answer?

Evidence can be misread or ignored

The paper's analysis of SpatialClaw finds failures even when the tools obtain correct geometric evidence.

An agent may reverse a reference frame, discard a useful reading, or rely on visual intuition when forming its final answer.

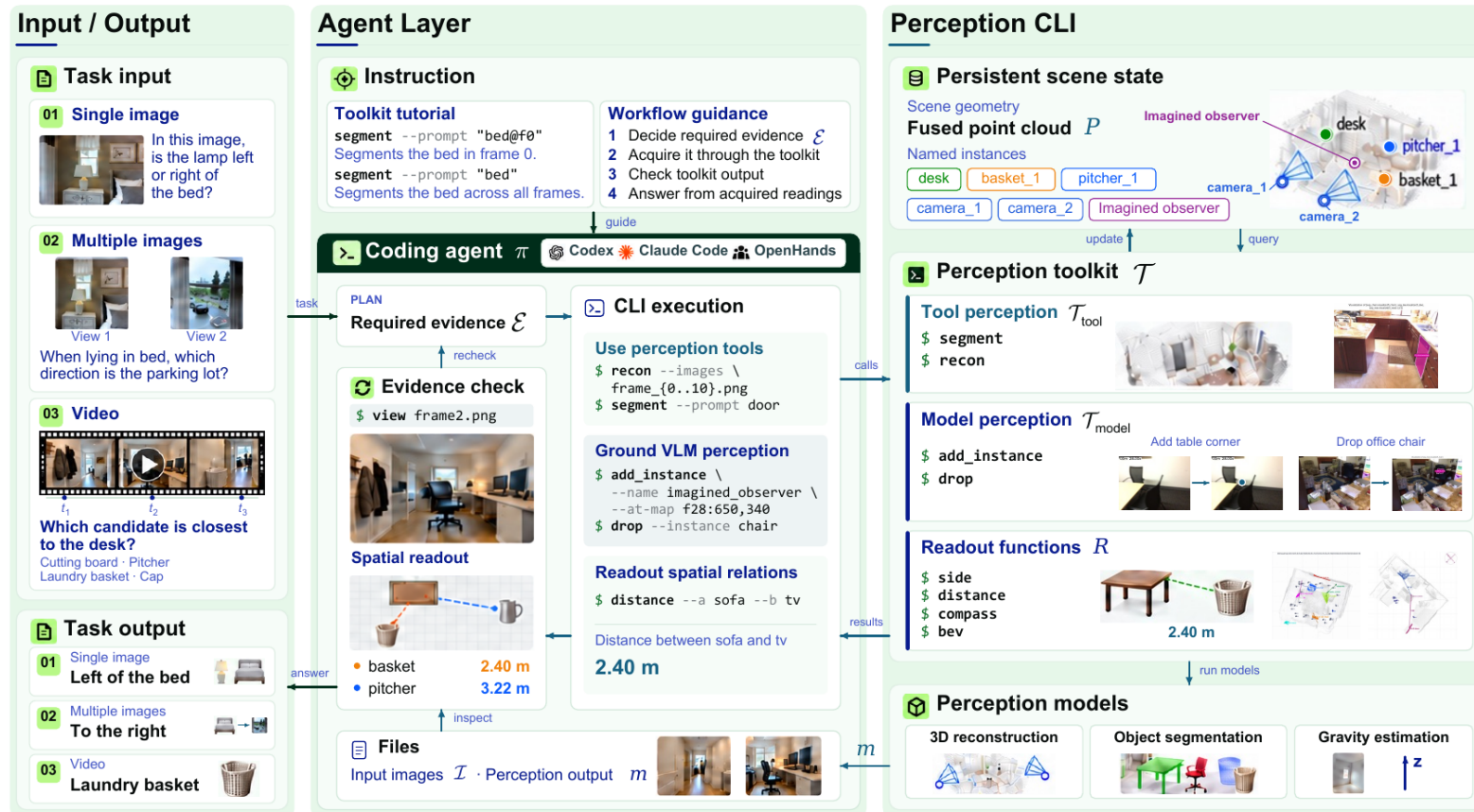
Keep answers grounded in evidence

A general coding agent already provides the ability to inspect outputs and revise them. Equip it with a spatial perception interface.

Store evidence in scene memory, ground the model's observations in that scene, and return interpreted spatial readings.

Key Idea: A General Agent with Spatial Grounding

A general coding agent uses scene memory and explicitly interpreted measurements.



General coding agent

The agent calls perception through its shell and inspects the resulting files.

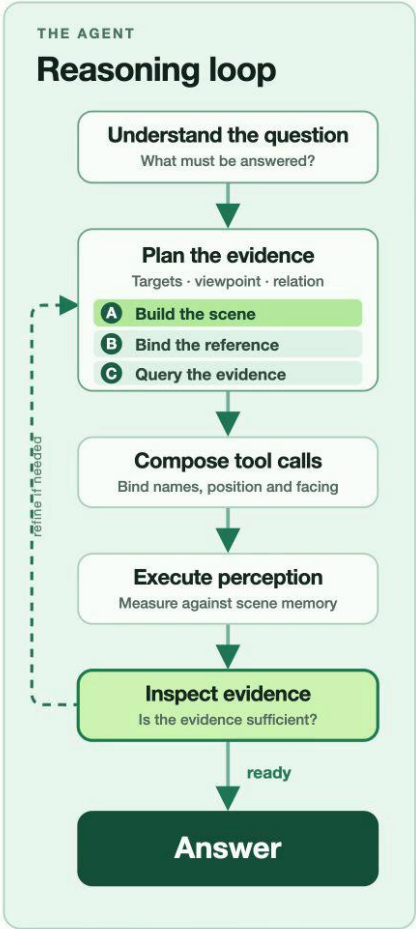
Shared scene memory

Model and tool perception update one named 3D scene that persists across steps.

Interpreted measurements

Readouts pair values and units with spatial conclusions in an explicit reference frame.

Demo: S-Agent 2.0



MULTI-VIEW SPATIAL REASONING

When lying in bed, where is the parking lot relative to you?

04

LOCATE THE TARGETS

Locate the parking lot, bed and headboard.

INSPECT

Spatial CLI

perception.py

COMMANDS · KEY ARGUMENTS

```
$ segment --image frame_0.png \
--frame-index 0 \
--prompt "parking lot"

$ segment --image frame_1.png \
--frame-index 1 \
--prompt "bed" --prompt "headboard"
```

↳ 3 named objects stored

f0:0 parking lot
f1:0 bed f1:1 headboard

Segmentation updates the existing scene.

Object evidence

3 named instances



f0 Parking lot



f1 Bed + headboard

parking lot

bed

headboard

Image masks bind the object names to visual evidence.



PERSISTENT SCENE STATE

parking lot · bed · headboard

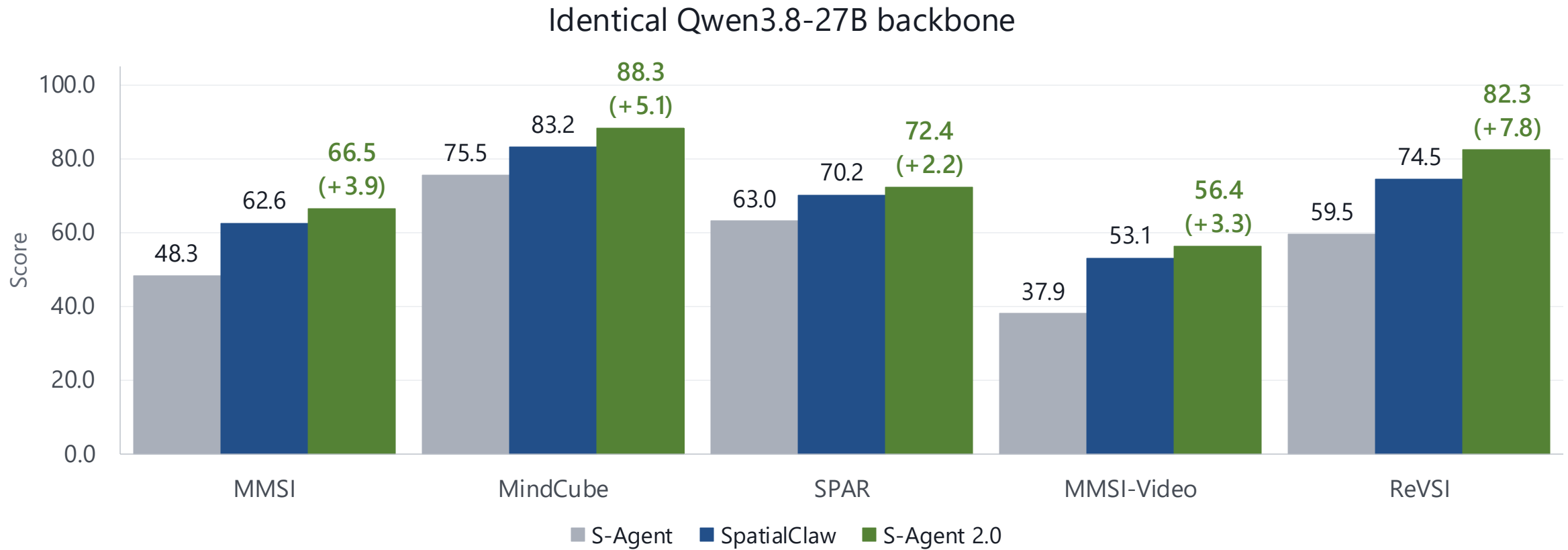
3 objects

Recorded CLI outputs · presentation-paced replay

00:05 / 00:10

Click to play full screen (15 s)

Results: Consistent Gains Across Five Benchmarks



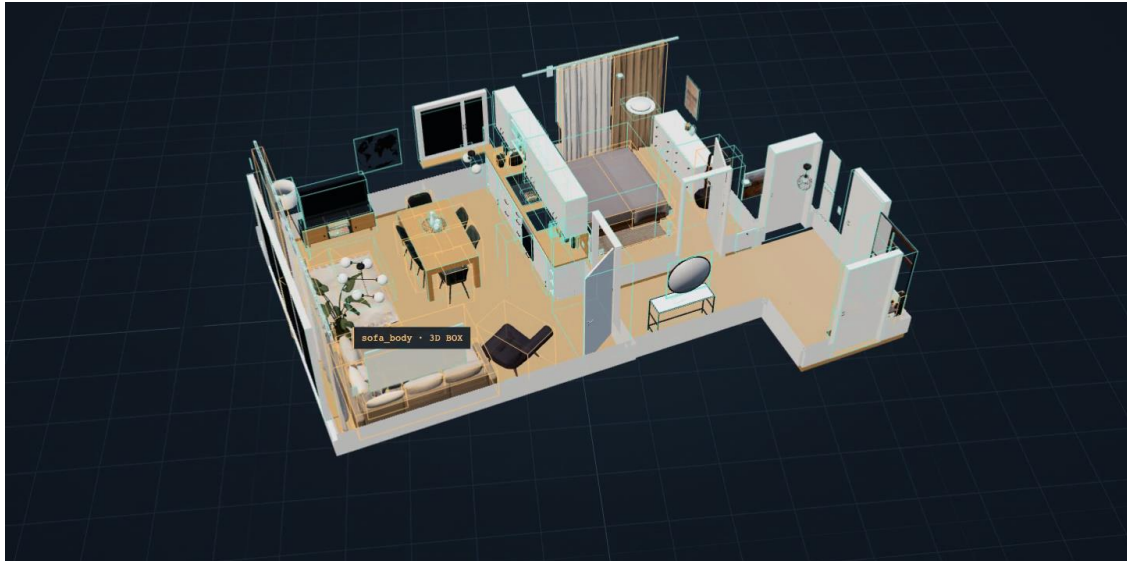
5 / 5

benchmarks improve over SpatialClaw
Parentheses show the gain in score points

Further Examples: Agentic Scene Generation

Object-level scene generation at apartment scale

Apartment A: 3D build process



5000+ editable scene objects

Apartment B: Continuous scene tour

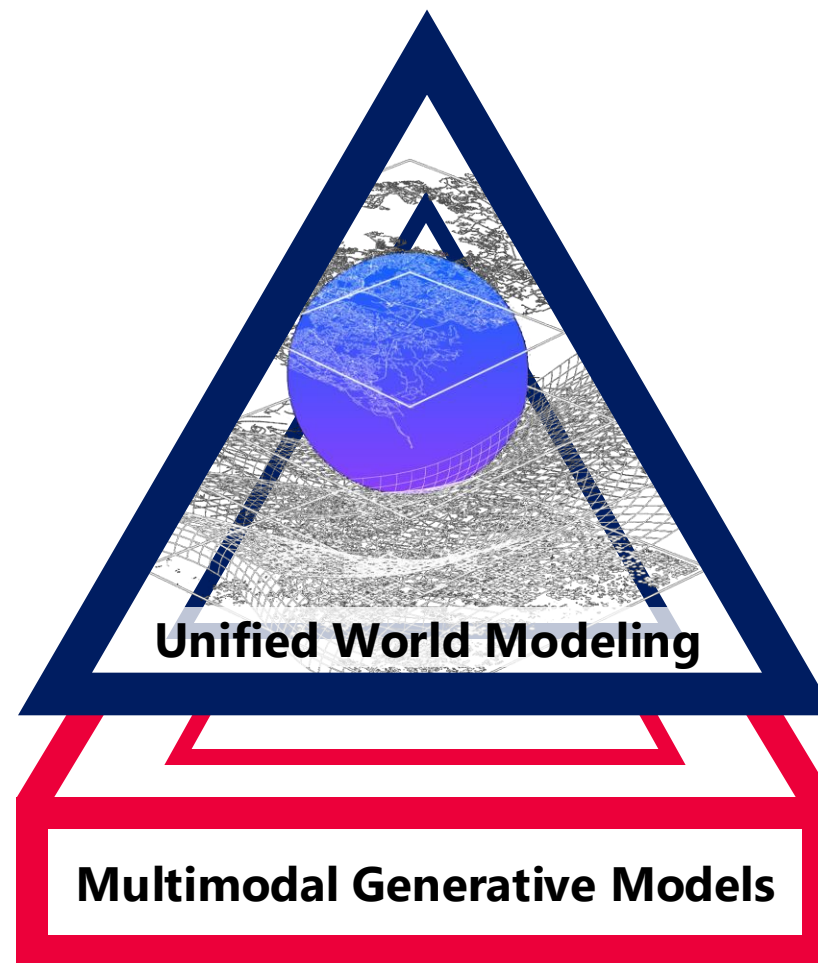


500+ editable scene objects

Be Physical

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Be Actionable

How to Interact with
the Physical World

Be Actionable: Xperience-0

Xperience-0: Scaling Loco-Manipulation World Models across Embodiments with Human Experience

Tianqi Liu, Hanfei Zhao, Qifeng Liang, Hui En Pang, Jiarui Xu, Sayang Mu, Penghao Qiu, YingSheng Zhang,

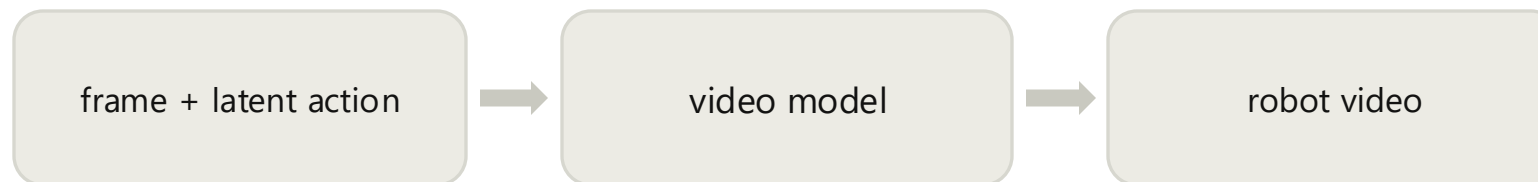
Jun Wang, Ziang Cao, Mingze Sun, Yukang Cao, Fangzhou Hong, Zhaoxi Chen, Ziwei Liu

Challenges

Embodiment Heterogeneity · Loco-Manipulation Complexity · Scarce Multimodal Supervision

DreamDojo

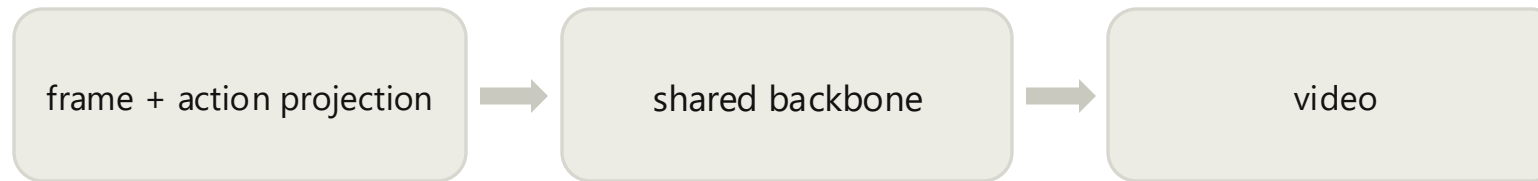
NVIDIA · 2026



- per-embodiment latent actions

Cosmos-3

NVIDIA · 2026



- domain-specific projections

AnyWorld

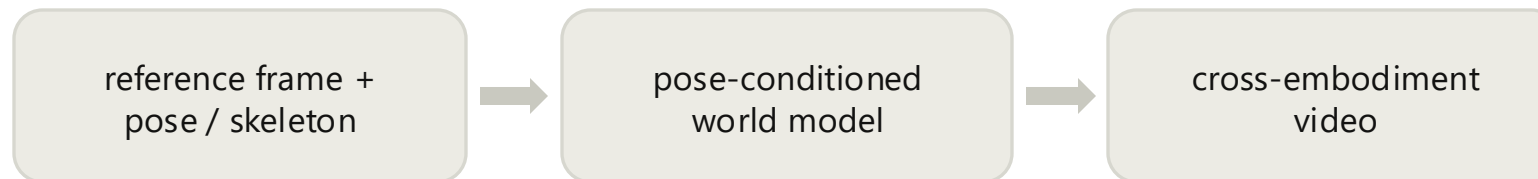
XPENG · 2026

OSCAR

NVIDIA · 2026

RynnWorld-Teleop

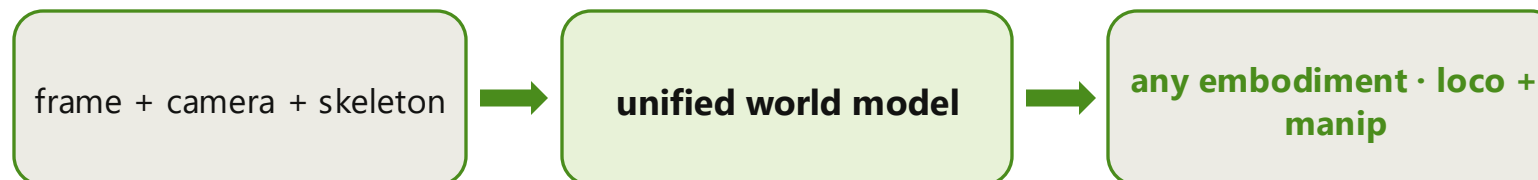
Alibaba DAMO · 2026



manipulation-focused applications

Xperience-0

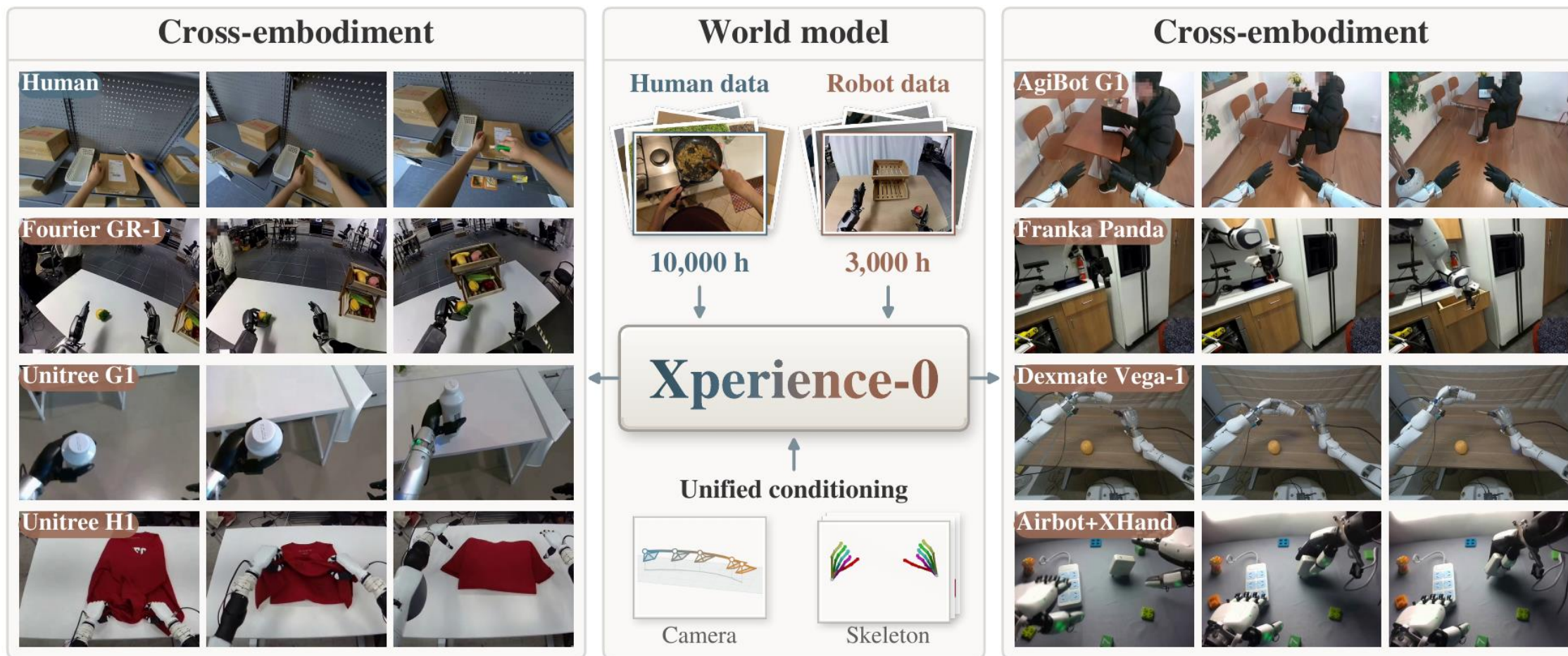
Ropedia · 2026 · OURS



+ one interface fits all bodies

Xperience-0 · Overview

A cross-embodiment loco-manipulation world model learned from human experience



Key Idea · Xperience Data

10,000 h human video + 3,000 h robot data, one unified annotation format.



Human video

Robot data

Camera trajectories · skeletons · captions · human-in-the-loop QC · + Xp-Teleop long-horizon benchmark

Key Idea · Unified Representation

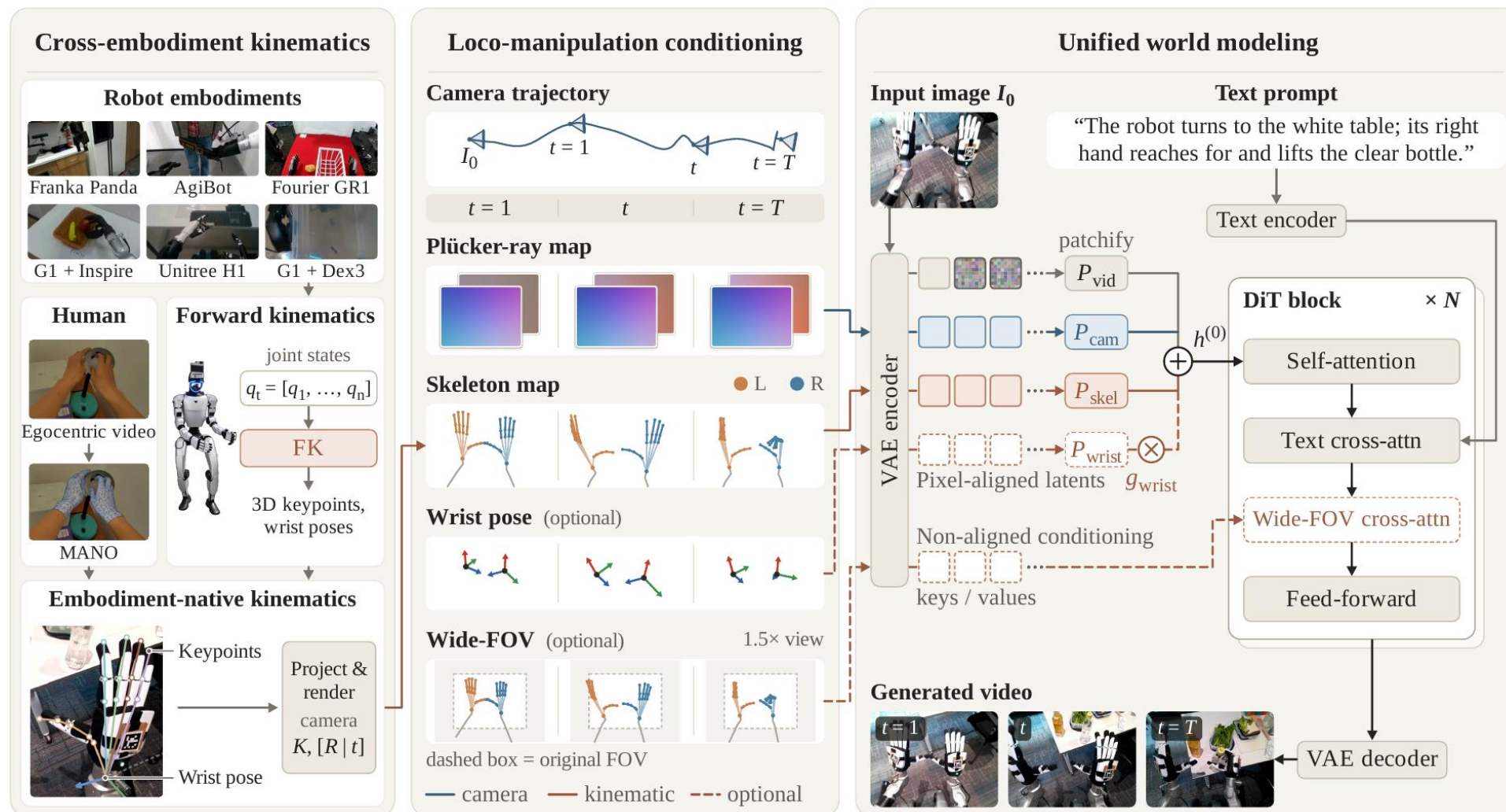
Camera poses + projected skeletons -
one interface for all embodiments.

- Camera -> walking / viewpoint
- Skeleton -> hands / manipulation
- Humans, grippers, dexterous hands



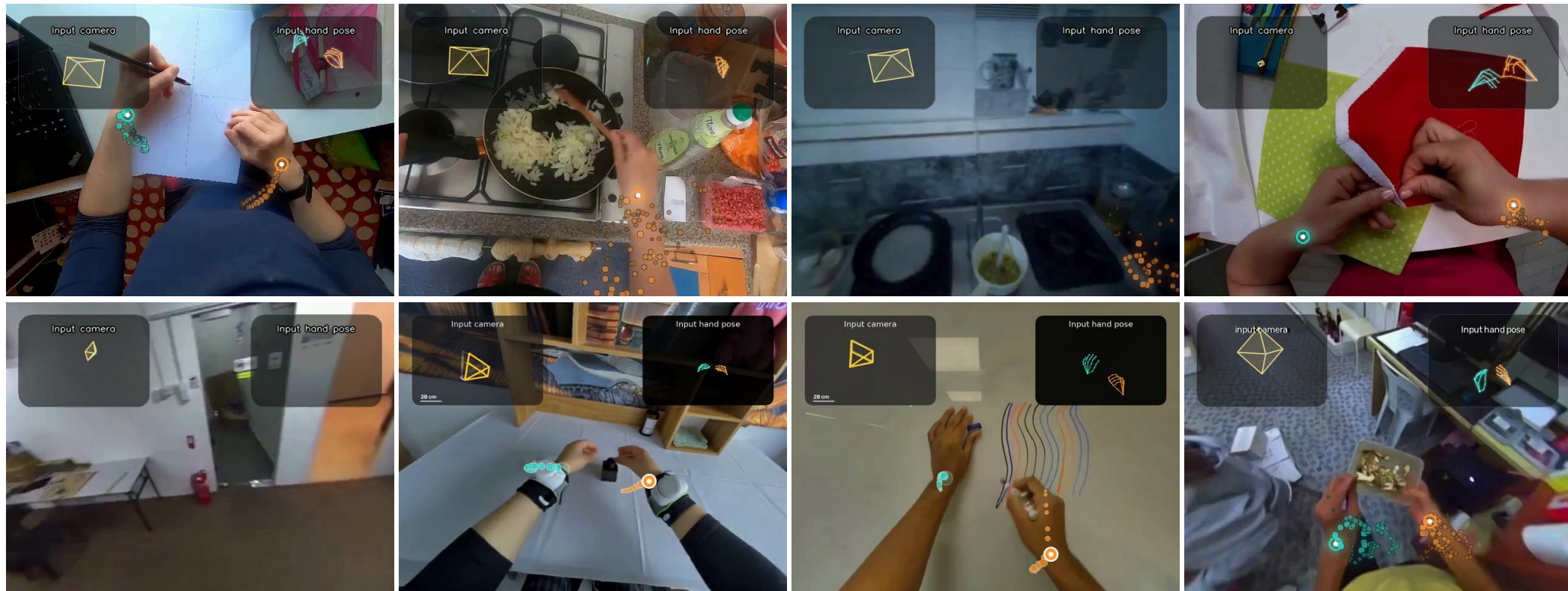
Key Idea · Unified World Model

Camera poses + projected skeletons → a video world model; one interface for every embodiment



Human Video Pre-training

Learning **Locomotion** and **Manipulation** Dynamics from **10K Hours** of Egocentric Human Videos

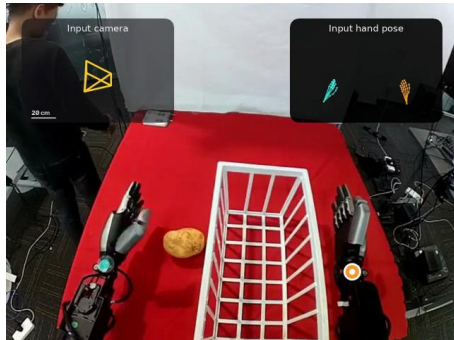


Results are generated by our model across diverse unseen scenes.

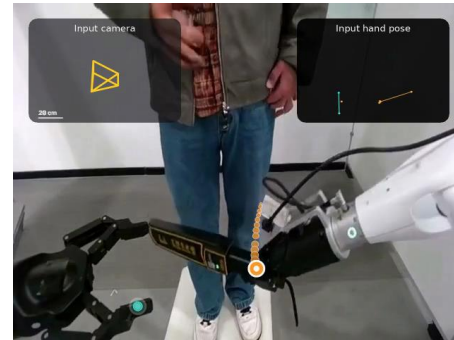
From Human Experience to Multi-Embodiment Robots

One Model, Multiple Embodiments

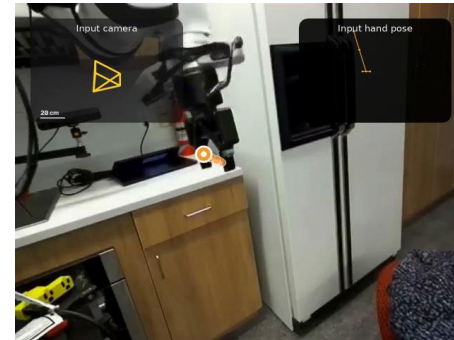
Fourier



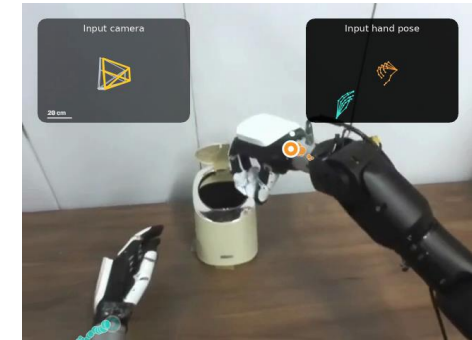
AgiBotWorld



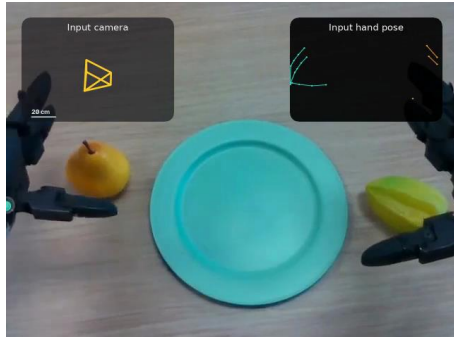
DROID



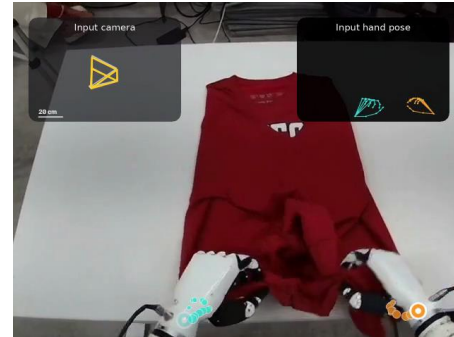
DECO50



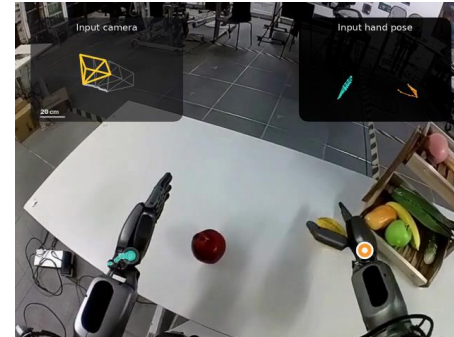
GR00T-G1



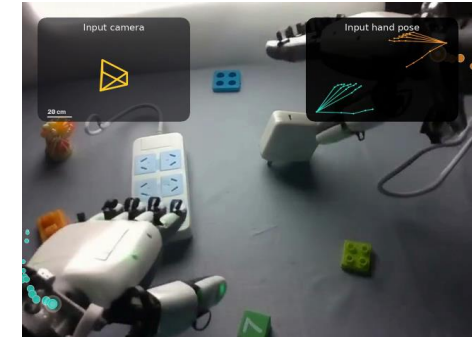
HumanPlus



GR1



Dexora



Zero-shot generation on held-out clips across diverse robot embodiments.

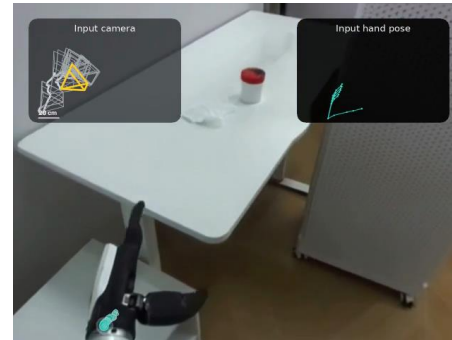
Long-Horizon Rollout

Continuous autoregressive generation with locomotion and manipulation (30 s rollouts)

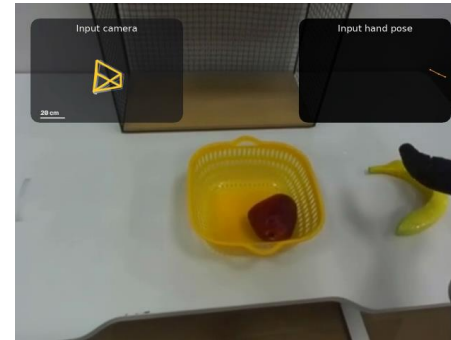
Fridge Interaction



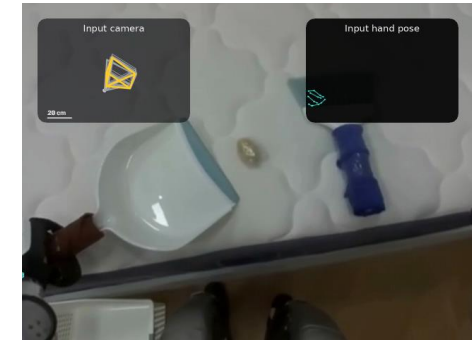
Garbage Collection



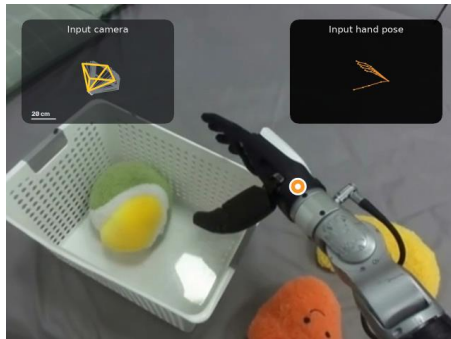
Fruit Arrangement



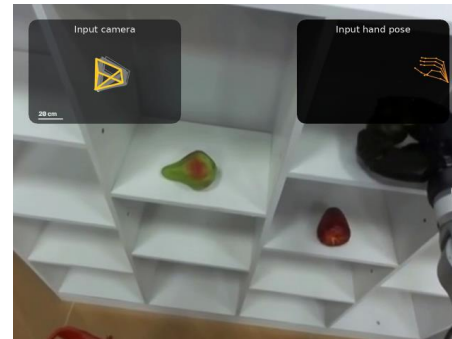
Bed Tidying



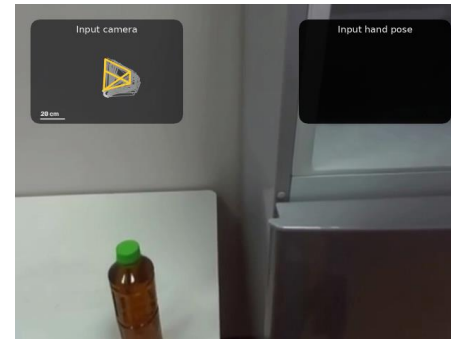
Toy Collection



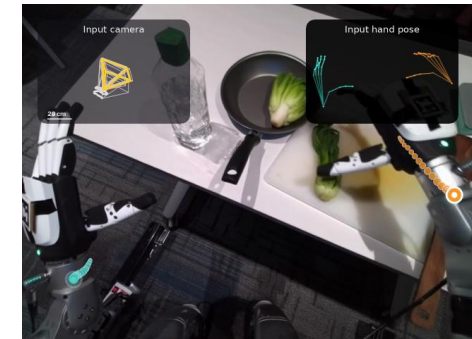
Closet Arrangement



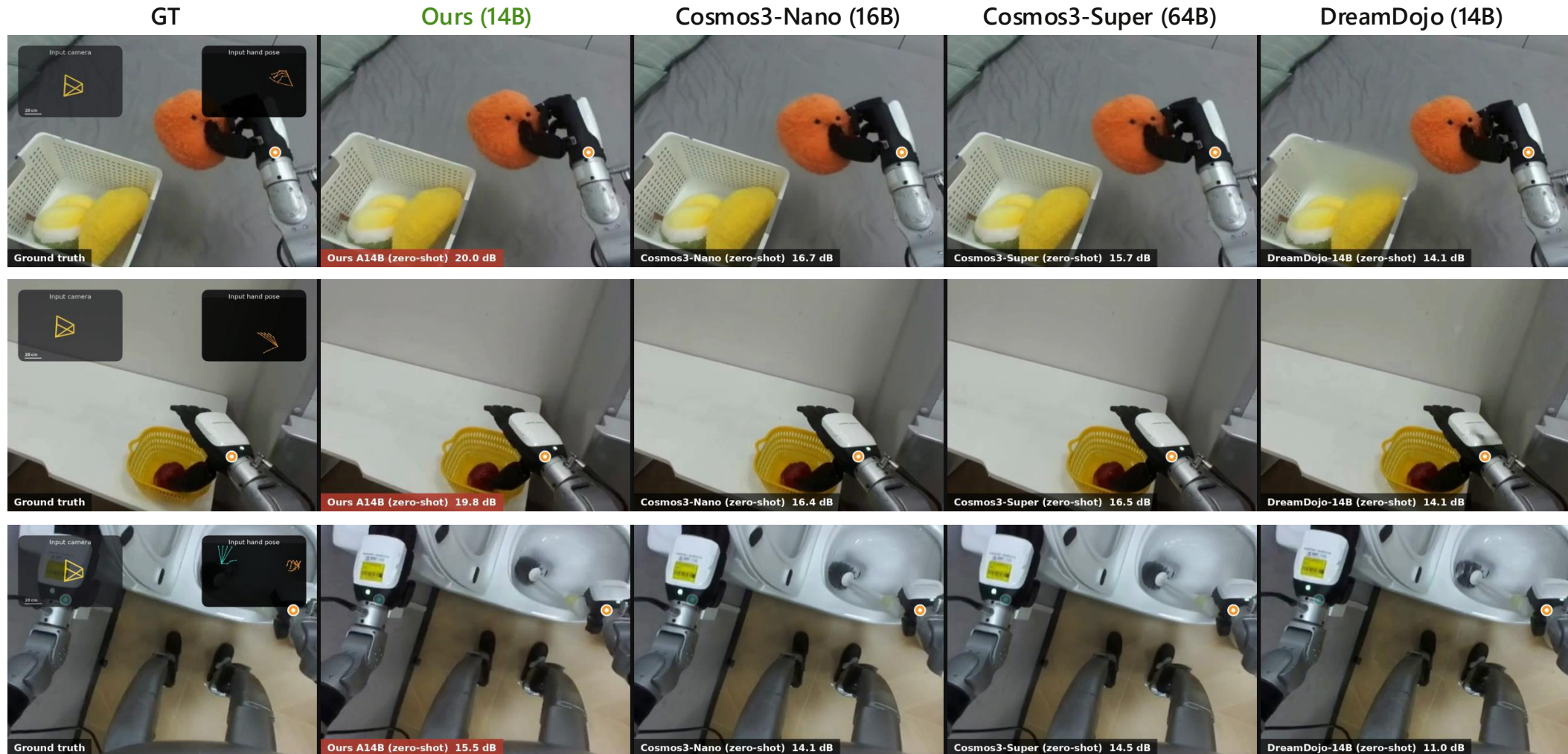
Upper-Fridge Interaction



Teleoperated Manipulation

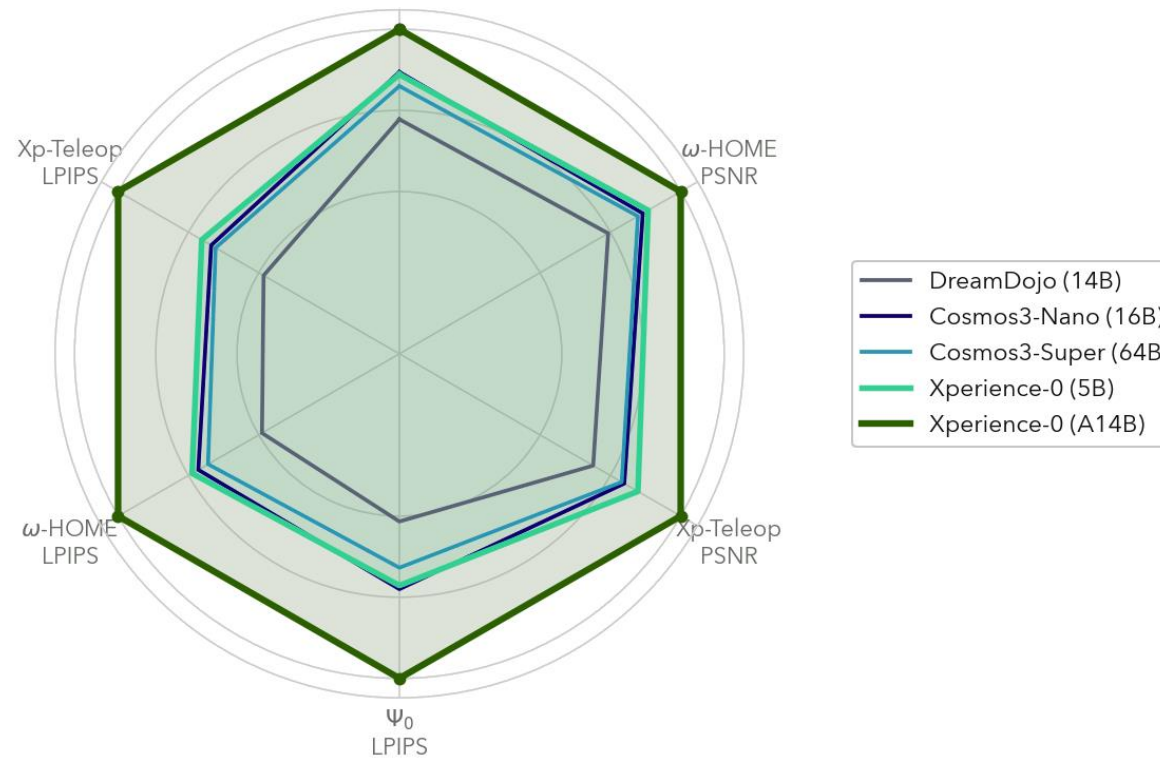


Comparison with State-of-the-Art Methods



Experiment Results · Benchmarks

Xperience-0 (A14B) and (5B) achieve first- and second-best performance across many different benchmarks.



full-data; each axis normalized so the leader = 1.0 (LPIPS inverted: lower is better)

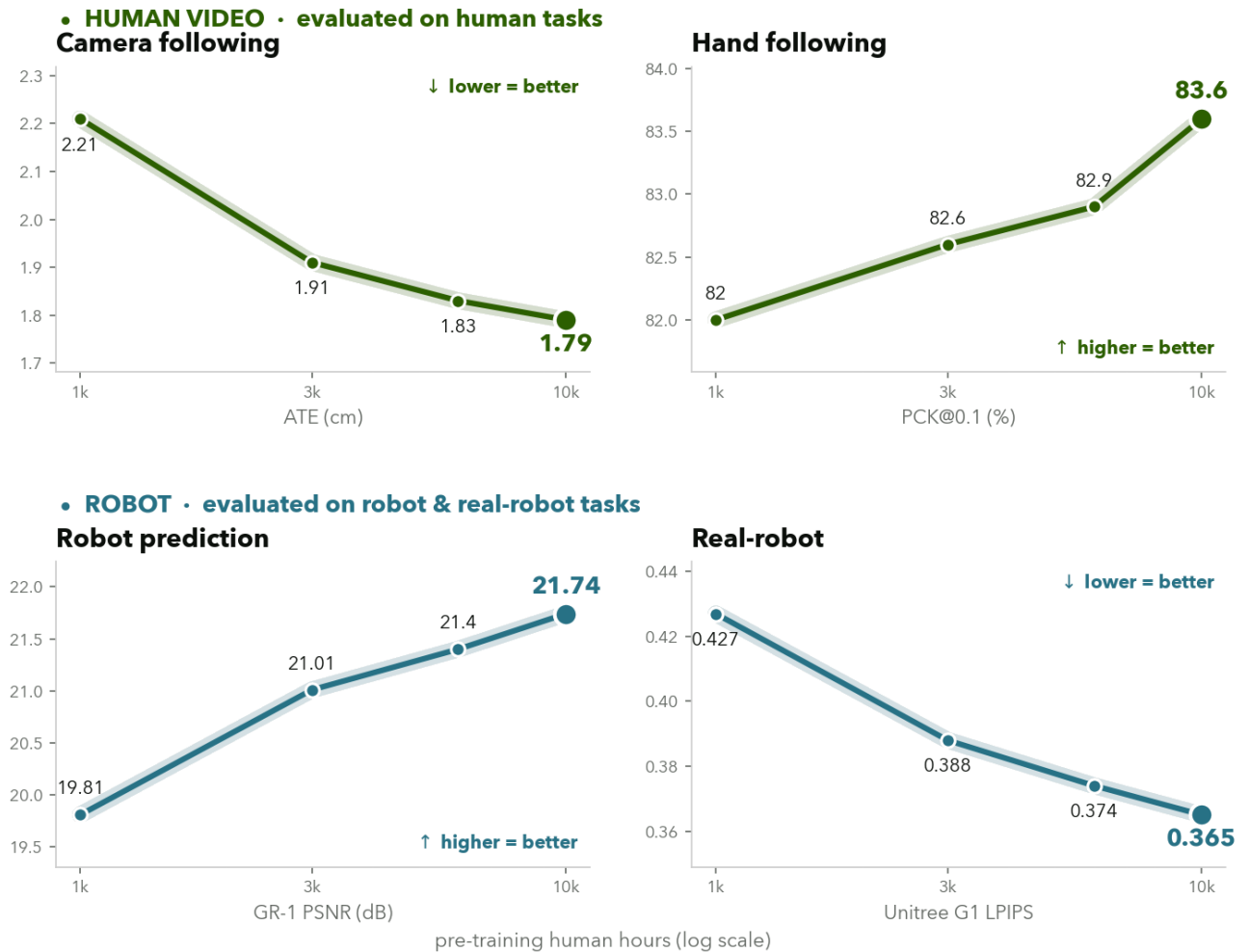
Scaling with human experience

More human video ->

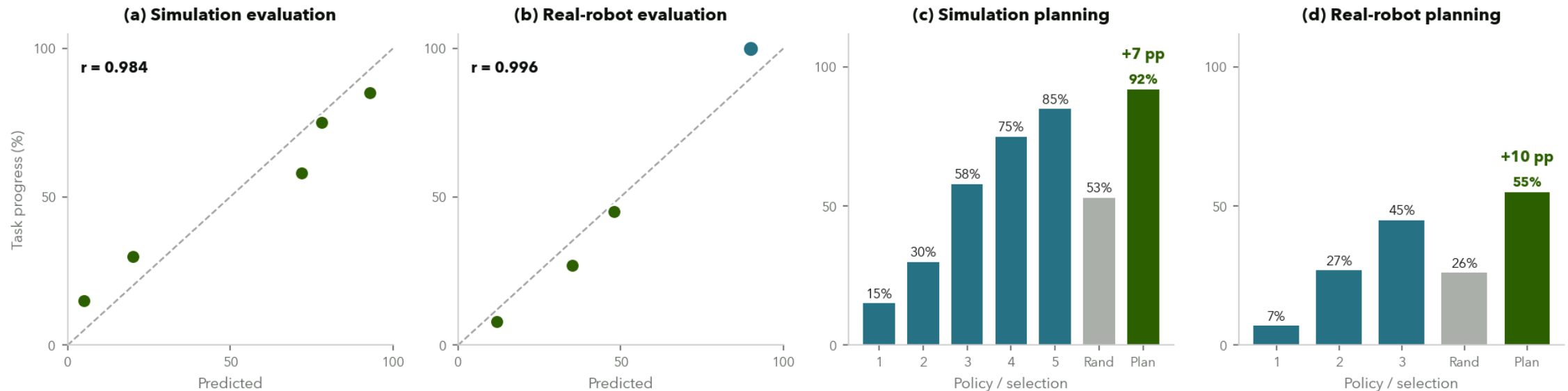
a better world model, every metric.

— Camera, hand, robot, real-robot all improve

More human-video pre-training -> better world model



Downstream results



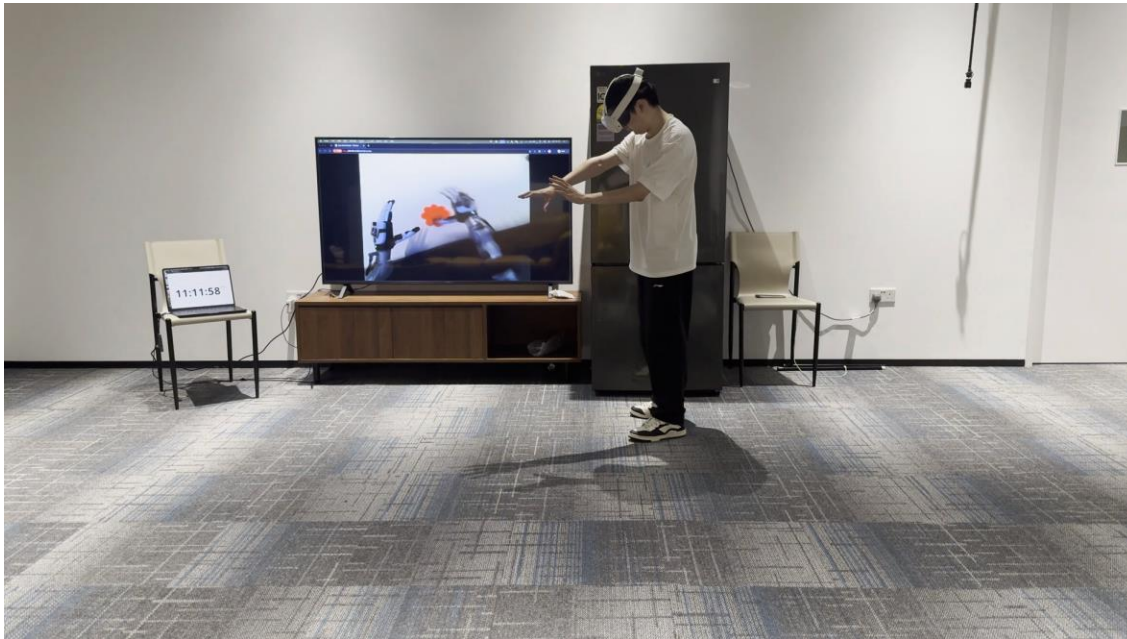
Policy evaluation matches reality ($r = 0.984 / 0.996$) · planning 92% vs 53% (sim), 55% vs 26% (real)

Policy rankings agree closely with real outcomes in the tested settings.

Real-Time Interactive Control

External recording of the operator; the display shows the ego view.

01 Robot-hand interaction



Object grasping and movement
with robot hands in the displayed ego view.

02 Human-hand interaction



Viewpoint changes and tabletop object
interaction in the displayed ego view.

Real-Robot Demo

Ego: robot view Exo: external view Task: pick up the object and place it in the container

01 Octopus toy / Ego



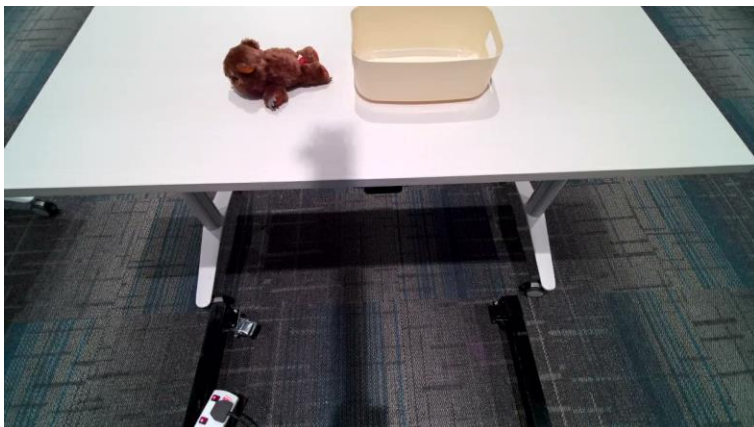
01 Octopus toy / Exo



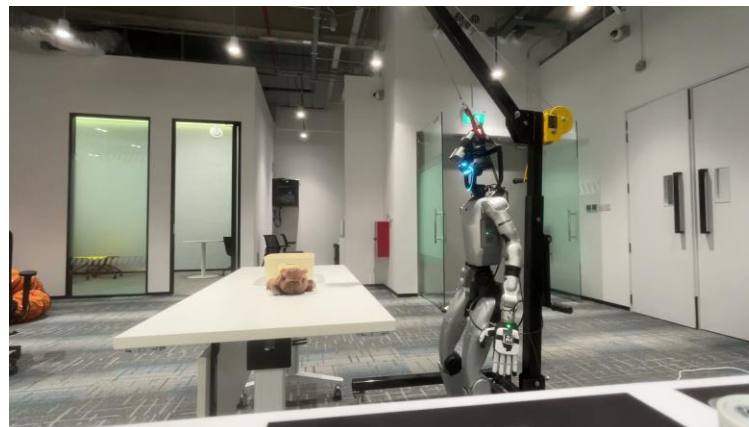
03 Snack bag / Ego



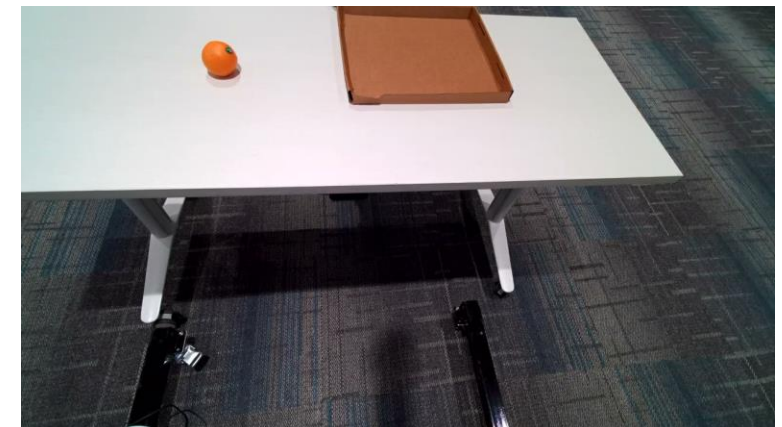
02 Plush toy / Ego



02 Plush toy / Exo



04 Orange / Ego



Real-Robot Demo

Each video pairs an external view (Exo, left) with the robot view (Ego, right).

01 Tabletop wiping and collection

Walk over to the table, pick up the cloth, wipe the tabletop, then place it in the basket.



02 Bedside collection

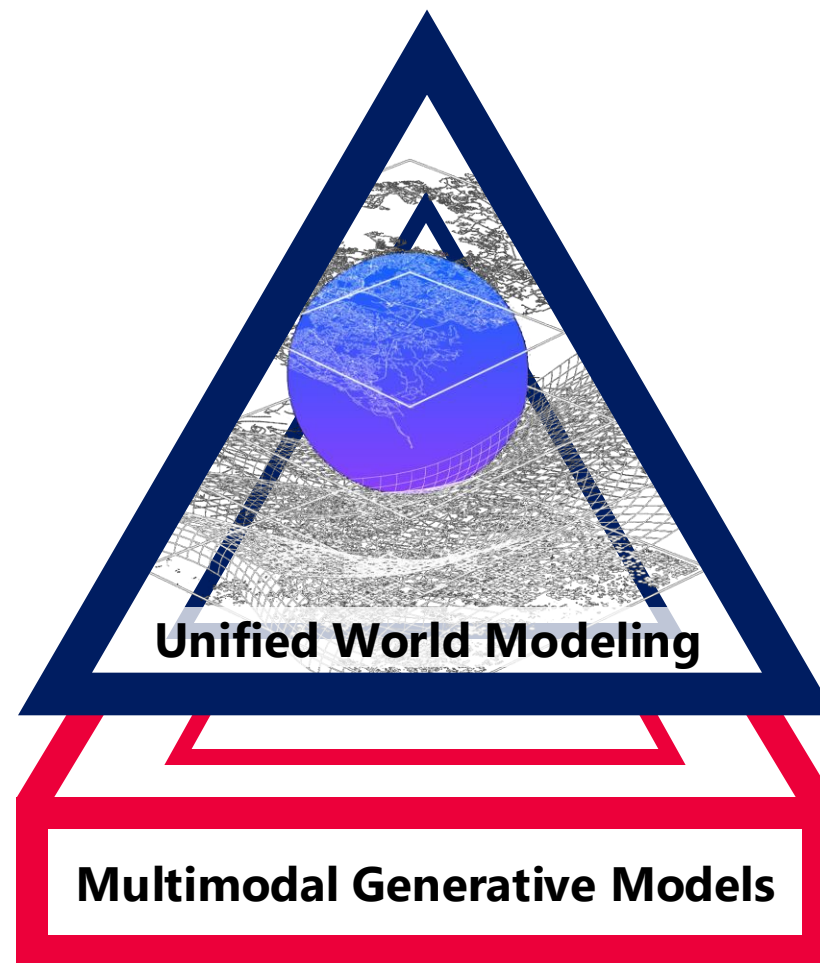
Pick up the cloth from the bed and place it in the bag.



Be Physical

How to Model Physical Properties

Be Agentic
How to Understand
the Physical World



Be Actionable
How to Interact with
the Physical World



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INTELLIGENCE

Thank You

Ziwei Liu 刘子纬

Nanyang Technological University

